Pigeonhole principles in Ramsey theory

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Introduction

When one learns about Ramsey theory, one starts to realize a common structure of many theorems. There is usually a pigeonhole theorem, and then a Ramsey-style theorem derived from this pigeonhole. Table 1 lists these correspondences, and they are what we will study.

Pigeonhole principle	**Derived Ramsey theorem**
Dirichlet's principle	Ramsey's theorem
Hindman's theorem	Milliken–Taylor theorem
Hales–Jewett theorem	Graham–Rothschild theorem
Carlson–Simpson lemma	ω-dimensional Graham–Rothschild
Halpern–Läuchli theorem	Milliken's tree theorem

Table 1: Pigeonhole principles and the Ramsey-style theorems derived from them.

Interestingly, proving the pigeonhole usually turns out to be the hard part, and the Ramsey-style theorem then automatically follows by a standard fusion argument.

Another recurring theme in Ramsey theory is that some classical "linear" partition theorems admit natural analogues for trees. The linear theorem then corresponds to the simplest possible tree consisting of a single branch. We will study these tree theorems as well.

Linear theorem	**Tree analogue**
Dirichlet's principle	Halpern–Läuchli theorem
Ramsey's theorem	Milliken's tree theorem
Graham–Rothschild theorem	Todorčević–Tyros theorem

Table 2: Linear Ramsey theorems and their tree analogues.

1 Preliminaries

We will work within the Zermelo–Fraenkel set theory ZF and we will usually assume the axiom of countable choice, the statement that every countable set x has a *choice function*: a function f with domain x such that for every nonempty set $t \in x$ we have $f(t) \in t$.

We assume basic knowledge of ordinals, cardinals, and ultrafilters, and we include a brief overview of these topics in the Appendix.

Set-theoretic notation When we say that there is a 1-*to*-1 *correspondence* between two sets A and B, we mean that there exists a bijection between them. The power set of a set X is denoted by $\mathcal{P}(X)$, the set of all subsets of X that have cardinality λ is denoted by $[X]^\lambda$, and the set of all subsets of X that have cardinality $< \lambda$ is denoted by $[X]^{<\lambda}$.

We use the set-theoretic convention that a natural number n is the set $\{0, 1, \dots, n-1\}$ and that the set of all natural numbers is denoted by ω. Moreover, $n < \omega$ and $n \leq \omega$ mean $n \in \omega$ and $n \in \omega \cup \{\omega\}$, respectively.

Dirichlet's principle Dirichlet's principle is the fancy name for the usual infinite version of the pigeonhole principle. It states that if the union of finitely many sets $A_0, A_1, \dots, A_n$ is infinite, then one of the sets A_i had to be infinite.

Order-theoretic trees A *tree* is a partially ordered set $(T, <_T)$ such that for every $x \in T$, the set

$$(\leftarrow, x) \coloneqq \{y \in T \mid y <_T x\}$$

is a finite chain, and there is a unique minimal element called the *root* of T.

The elements $x \in T$ of a tree are called the *nodes* of T. In graph-theoretic terms, $y <_T x$ means that y lies on the unique path $(\leftarrow, x)$ from x to the root.

A *subtree* of a tree $(T, <_T)$ is any subset $S \subseteq T$ together with the inherited order that is also a tree (has a unique minimal element). We will often simply write T instead of $(T, <_T)$ and $<$ instead of $<_T$ if the order is clear from the context. Two trees T and S are *isomorphic* if they are order-isomorphic as partially ordered sets.

The subtree *sprouting* from $x \in T$ is the subtree $T_x \coloneqq \{x' \in T \mid x \leq x'\}$. A node y is called a *predecessor* of x if $y < x$, a *successor* of x if $x < y$, and an *immediate successor* of x if it is a $\leq$-minimal successor of x. By nodes *below* and *above* x, we mean the set of predecessors and successors of x, respectively. A node x is called a *leaf* if it has no immediate successors. The *branching factor* of x is the cardinality of the set of its immediate successors. If every node has finite branching factor, then we say that the tree is *finitely branching*.

The *height* or *level* of x is $|x|_T \coloneqq |(\leftarrow, x)|$. A tree is divided into *levels*, starting with level zero, which contains the root. The *n-th level* of T is the set

$$T(n) \coloneqq \{x \in T \mid |x|_T = n\}.$$

The *height* $\mathrm{H}(T)$ of a tree T is the least n such that $T(n) = \varnothing$, and if no such finite n exists, then $\mathrm{H}(T)$ is the first infinite ordinal number ω.

A *branch* of a tree is a $\subseteq$-maximal chain. The *length* of a branch B is $|B|$. It is easy to see that every branch has length at most $\mathrm{H}(T)$. A *cofinal branch* is a branch of length $\mathrm{H}(T)$.

König's lemma

Lemma 1.1 (König's lemma)**.** *Every infinite, finitely branching tree T has an infinite branch.*

Proof. We say that a node $x \in T$ is good if the subtree sprouting from it T_x is infinite. Because T is infinite, its root r is good. Because r has finite branching factor, one of its immediate successors x_0 has to be good (from Dirichlet's principle). Then we choose x_1 as a good immediate successor of x_0 and so on. We construct an infinite sequence of nodes $(x_n)_{n<\omega}$ such that x_{n+1} is an immediate successor of x_n, and T_{x_n} is infinite for all n. This is a branch of length ω. □

Remark. Since we are *choosing* x_{n+1} from the set of good immediate successors of x_n for infinitely many $n \in \omega$, we are using some kind of choice principle. König's lemma is in fact equivalent to the axiom of countable choice for finite sets. Briefly, countable choice for finite sets implies that a countable union of finite sets is countable. Since every level in a finitely branching tree is finite, it follows that the entire tree is countable. Therefore we can well-order it via a bijection with ω. Finally, It is fairly easy to derive countable choice for finite sets from König's lemma, one basically uses the same argument we will soon use to prove the compactness principle.

Hypergraphs A *hypergraph* is a pair $H = (V, E)$ where V is a set of *vertices* and $E \subseteq [V]^{<\omega}$ is a set of *hyperedges* (finite sets of vertices). For $W \subseteq V$, we denote by H_W the *induced hypergraph* $(W, E \cap \mathcal{P}(W))$. An *$r$-coloring* of H is a function $\chi\colon V \to r$. We say that a hyperedge $e \in E$ is *monochromatic* with respect to χ if $f \restriction e$ is constant (all $v \in e$ have the same color). An r-coloring of H is *proper* if no hyperedge $e \in E$ is monochromatic. H is *r-colorable* if it has a proper r-coloring. The *chromatic number* $\chi(H)$ of H is the least r such that H is r-colorable.

Compactness Principle There are many statements in mathematics that go under the name of "compactness principle." Their common property is the ability to transfer statements about finite objects to statements about infinite objects, and the fact that the axiom of choice is required to prove them.

The following version of the compactness principle is particularly useful in Ramsey theory, as it allows us to easily convert many infinite Ramsey-style theorems into their finite versions.

Theorem 1.2 (Compactness principle)**.** *Let $H = (V, E)$ be a hypergraph with countable V and $E \subseteq [V]^{<\omega}$. If $\chi(H) > r$, then there exists some finite $W \subseteq V$ such that $\chi(H_W) > r$.*

Remark. This theorem is true even when V is uncountable and it is then equivalent to the Ultrafilter lemma, another weak form of the axiom of choice. Try deriving the full version of this theorem (when V can be uncountable) from Zorn's lemma.

Proof. We will show that if every finite $W \subseteq V$ satisfies $\chi(H_W) \leq r$, then $\chi(H) \leq r$. Since V is countable, we can enumerate it as $V = \{v_0, v_1, v_2, \dots\}$. Construct a tree whose nodes are finite r-colorings $f\colon W \to r$ where $W = \{v_0, v_1, \dots, v_n\}$ for some n, and f is below g in the tree order if g extends f. This tree has height ω and is finitely branching, so there exists a cofinal branch, which corresponds to an $r$coloring of the entire graph. □

Partition arrow notation For cardinals κ, λ, μ and a natural number $k \in \omega$, the expression

$$\kappa \longrightarrow (\lambda)^k_\mu$$

means that for every mapping $\chi : [\kappa]^k \to \mu$, there exists $A \subseteq \kappa$ of cardinality λ such that $\chi \restriction [A]^k$ is constant. We say that χ is a *coloring* of *k-sets* of κ by μ colors, and that A is *homogeneous* for χ. We also say that the set $[A]^k$ is *monochromatic* with respect to the coloring χ.

We read the expression $\kappa \to (\lambda)^k_\mu$ as "κ arrows[1] k-sets of λ with μ colors."

2 Ramsey's Theorem

We begin with the classical theorem of Frank Ramsey [Ram30] which started it all.

2.1 Infinite Ramsey Theorem

Theorem 2.1 (Infinite Ramsey Theorem)**.** *For every number of colors r and dimension k we have $\omega \longrightarrow (\omega)^k_r$.*

Written fully: For every number of colors r, dimension k, and for every r-coloring $\chi\colon [\omega]^k \to r$ of k-sets of ω, there exists an infinite subset $A \subseteq \omega$ such that $\chi \restriction [A]^k$ is constant (every k-set of A has the same color). We say that A is *homogeneous* for χ.

Proof. We proceed by induction on k. The base case $k = 1$ is Dirichlet's principle. Suppose $k \geq 2$ and χ is a coloring of k-sets of ω. For $t \in \omega$ define a coloring of $(k-1)$-sets of $\omega \setminus \{t\}$ as

$$\chi_t(A) \coloneqq \chi(A \cup \{t\}).$$

Let $t_0 \coloneqq 0$ and consider the coloring χ_{t_0} of $[\omega \setminus \{t_0\}]^{k-1}$. Use the induction hypothesis for $k-1$ to obtain an infinite $A_0 \subseteq \omega \setminus \{t_0\}$ homogeneous for χ_{t_0}. Notice that every k-set of $\{t_0\} \cup A_0$ that contains t_0 has the same color c_0 with respect to χ.

We repeat this process. Let $t_1 \coloneqq \min(A_0)$ and consider the coloring χ_{t_1} of $[A_0 \setminus \{t_1\}]^{k-1}$. By using the induction hypothesis, we obtain an infinite $A_1 \subseteq A_0 \setminus \{t_1\}$ homogeneous for χ_{t_1}.

[1] Česky říkáme „šipkuje."

Note that every k-set of $\{t_0\} \cup \{t_1\} \cup A_1$ that contains t_1 has the same color c_1, which might differ from c_0.

Repeating this process indefinitely, we construct an infinite sequence $t_0, t_1, \ldots$ such that for every t_n it holds that every k-set of $\{t_i \mid i < \omega\}$ containing t_n has the same color c_n. Since there are only finitely many colors, by Dirichlet's principle, there exists an infinite subsequence $t_{i_0}, t_{i_1}, t_{i_2}, \ldots$ such that all t_{i_j} correspond to the same color. Hence, the set $\{t_{i_j} \mid j < \omega\}$ is homogeneous for χ. □

Ramsey's theorem implies that every infinite graph either contains an infinite independent set or an infinite clique. Indeed, take a countable subset of vertices and color pairs of them red if they are connected by an edge, and blue otherwise. By Ramsey's theorem, there is an infinite homogeneous subset of vertices.

Weakly compact cardinals In particular, every graph on ω vertices contains a clique of size ω or an independent set of size ω. Note that ZFC cannot prove the existence of another cardinal with this property — a cardinal $\kappa > \omega$ such that every graph on κ vertices contains a clique of size κ or an independent set of size κ. Such a cardinal would be called *weakly compact*. Weakly compact cardinals are large cardinals (see Appendix A.2) much stronger than inaccessible cardinals, but still quite low in the large cardinal hierarchy.

Ramsey's theorem and the axiom of choice When proving Ramsey's theorem, we always chose the next element t_{n+1} as the minimum of the remaining infinite subset A_n. We were able to do this because ω is well-ordered, but for a general infinite starting set, we would be invoking some weak form of the axiom of choice.

Theorem 2.2 (Kleinbern [Kle69])**.** *In* ZF*, it is impossible to prove that for every infinite set X and for every finite coloring $\chi\colon [X]^k \to r$ there exists an infinite $A \subseteq X$ homogeneous for χ.*

Theorem 2.3 (Lolli [Lol77])**.** *General Ramsey's theorem is equivalent to the axiom of countable choice for finite sets.*

Exercise 2.4. Prove from infinite Ramsey's theorem that

(a) Every infinite ordered set has an infinite chain or an infinite antichain.

(b) Every infinite linearly ordered set contains an infinite increasing sequence or an infinite decreasing sequence.

Note on the number of colors Before continuing, we make an important observation: It is usually enough to prove Ramsey-style theorems for coloring by two colors only since the general case then follows by induction.

Suppose that we have Ramsey's theorem for 2 colors and we would like r colors. Imagine the colors as blue and $r-1$ types of red. Ramsey's theorem for 2 colors gives us an infinite homogeneous subset. If it is blue, we are done. If it is a mix of reds, we repeat the process, noting that the number of colors has decreased.

The fusion argument Notice that in the proof above, when going from $k-1$ to k, we use the heavy machinery (the Induction Hypothesis for $k-1$) infinitely many times to generate the sequence $t_0, t_1, \ldots$, and then we use the simple tool (Dirichlet's principle) exactly once at the end to clean it up.

Now, we convert it into a "fusion" argument: we will use the simple tool (Dirichlet's principle) infinitely many times to "carve out" nested sets $A_n \subseteq \omega$ and then "fuse" them into a set $A = \{t_n \mid n < \omega\}$. This fused set A will be highly structured, allowing us to use the heavy machinery (the Induction Hypothesis) exactly once at the end.

Second proof of Ramsey's theorem. We proceed by induction on k. The base case $k = 1$ is Dirichlet's principle. Suppose $k \geq 2$ and χ is an r-coloring of k-sets of ω. We will construct a sequence of elements $t_0, t_1, t_2, \ldots$ and a sequence of infinite sets $\omega = A_0 \supseteq A_1 \supseteq A_2 \supseteq \ldots$ such that $t_n = \min(A_n)$. Let $A_0 \coloneqq \omega$ and $t_0 \coloneqq 0$.

Suppose that the set A_n and the element t_n have been defined. If $n < k - 2$ (so less than $k - 1$ sets have been defined), put $A_{n+1} \coloneqq A_n \setminus \{t_n\}$ and $t_{n+1} \coloneqq \min(A_{n+1})$.

Otherwise, consider the finite set of chosen elements $T_n \coloneqq \{t_0, t_1, \ldots, t_n\}$. There are only finitely many $(k-1)$-element subsets of T_n. For each such subset $S \in [T_n]^{k-1}$, we can view χ as coloring the elements $y \in A_n \setminus \{t_n\}$ according to the color of the k-set $S \cup \{y\}$. By applying Dirichlet's principle successively for each of the finitely many subsets $S \in [T_n]^{k-1}$, we can find an infinite set $A_{n+1} \subseteq A_n \setminus \{t_n\}$ such that for every $S \in [T_n]^{k-1}$, the color of $S \cup \{y\}$ is constant for all $y \in A_{n+1}$ (but it might differ for different S). Let c_S be the color corresponding to S. We then put $t_{n+1} \coloneqq \min(A_{n+1})$.

Repeating this process indefinitely, we obtain an infinite sequence of elements $t_0, t_1, \ldots$ forming a set $A \coloneqq \{t_n \mid n < \omega\}$. Observe that by our construction, for any k-set in A, its color under χ depends entirely on its first $k - 1$ elements. Define a new coloring χ' of $(k-1)$-sets of A by $\chi'(S) \coloneqq c_S$ for $S \in [A]^{k-1}$.

We now apply the induction hypothesis for $k - 1$ exactly once to the coloring χ' on $[A]^{k-1}$. This yields an infinite set $H \subseteq A$ that is homogeneous for χ'. Since all $(k-1)$-sets in H have the same color under χ', and since the χ-color of any k-set in H is dictated solely by its first $k - 1$ elements, it follows that H is homogeneous for the original coloring χ. □

Trick with signature In the fusion proof, we always applied Dirichlet's principle multiple times (more precisely, $\binom{n+1}{k-1}$ times) to obtain $A_{n+1} \subseteq A_n \setminus \{t_n\}$. However, we can do it in a single step using an important trick of looking at the *signature* of some object with respect to a given coloring function.

Third proof of Ramsey's theorem. Let a coloring $\chi\colon [\omega]^k \to 2$ be given. We repeat the same fusion argument as above, but stop at the point where we have $t_0, t_1, \ldots, t_n$ and the set A_n. To choose A_{n+1}, define for each $a \in A_n \setminus \{t_n\}$ its signature $\tau_a\colon [\{t_0, \ldots, t_n\}]^{k-1} \to \{0, 1\}$ by $\tau_a(S) \coloneqq \chi(S \cup \{a\})$. We define it as a function, but one may view τ_a as a vector of ones and zeroes of length $\binom{n+1}{k-1}$.

There are at most $2^{\binom{n+1}{k-1}}$ different signatures, so by Dirichlet's principle, there is an infinite subset $A_{n+1} \subseteq A_n \setminus \{t_n\}$ such that all $a \in A_{n+1}$ share the same signature τ. Notice that this accomplishes the same thing as when applying Dirichlet's principle $\binom{n+1}{k-1}$ times one by one. □

Remark. This proof implicitly hides the idea of the proof of König's lemma. When choosing which signature set to pick (into which subtree to descend), we always pick an infinite one.

Ultrafilter proof of Ramsey's theorem The last proof we include uses nonprinciple ultrafilters (see Appendix A.3). It will be similar to the fusion proof, but instead of Dirichlet's principle we will use a nonprinciple ultrafilter as an oracle. This approach is important because it can be generalized to prove that if κ is a weakly compact cardinal, meaning that $\kappa \longrightarrow (\kappa)^2_2$, then in fact $\kappa \longrightarrow (\kappa)^k_\mu$ for all $k \in \omega$ and $\mu < \kappa$.

Fourth proof of Ramsey's theorem. We again proceed by induction on k, with the base case being Dirichlet's principle. Suppose $k \geq 2$ and let $\chi\colon [\omega]^k \to r$ be given. Let $\mathcal{U} \subseteq \mathcal{P}(\omega)$ be a nonprinciple[2] ultrafilter on ω

[2]When proving this for a weakly compact cardinal κ, one needs something like a κ-complete uniform ultrafilter on every collection of κ many subsets of κ. Thankfully, every nonprinciple ultrafilter on ω is ω-complete and uniform, saving us a lot of work.

For every $S \in [\omega]^{k-1}$ and $i < r$ put

$$X_i(S) \coloneqq \{x \in \omega \setminus S \mid \chi(S \cup \{x\}) = i\}$$

The sets $X_i(S)$ for $i < r$ form a finite partition of the set $\omega \setminus S \in \mathcal{U}$, so there is a unique color $j < r$ such that $X_j(S) \in \mathcal{U}$. For a given S, denote this j by $j(S)$. Note that we have just defined a coloring

$$j \colon [\omega]^{k-1} \to r.$$

We will use it to construct an infinite set $Y = \{y_i \mid i < \omega\}$ that will contain the desired homogeneous set. We fix the first $k-1$ elements of Y as $y_i = 0$ for $i \leq k-2$. Suppose that we already have $Y_n = \{y_0, \ldots, y_n\}$ for some $n \geq k-2$. Note that the set $[Y_n]^{k-1}$ is finite and thus the set

$$A_n \coloneqq \bigcap \{X_{j(S)}(S) \mid S \in [Y_n]^{k-1}\}$$

is in $\mathcal{U}$ and is therefore infinite. Hence we can choose y_{n+1} as the least element of $A_n \setminus \{x \mid x \leq y_n\}$.

Now consider the restriction of the coloring j to the set $[Y]^{k-1}$. By the induction hypothesis, there is an infinite set $A \subseteq Y$ such that every $S \in [A]^{k-1}$ receives the same j-color c^*. We claim that A is homogeneous for χ with the same color c^*. Let $H \in [A]^k$. Put $y \coloneqq \max(H)$ and $S \coloneqq H \setminus \{y\}$. Then $j(S) = c^*$ and $y \in X_{c^*}(S)$ by the construction of Y. From the definition of the sets $X_i(S)$ we see that $\chi(H) = \chi(S \cup \{y\}) = c^*$. □

2.2 Finite Ramsey Theorem

We will now describe the general method of obtaining a finite Ramsey-style theorem from its infinite version, and we will demonstrate it on Ramsey's theorem.

Theorem 2.5 (Finite Ramsey Theorem)**.** *For every size n, number of colors r, and dimension k, there exists N such that $N \longrightarrow (n)^k_r$.*

Fully written: For every size n, number of colors r, and dimension k, there exists N such that for every r-coloring χ of $[N]^k$, there exists a subset $A \subseteq N$ of size n such that $\chi \restriction [A]^k$ is constant (every $a \in [A]^k$ has the same color).

In these compactness proofs, one always wants to construct a hypergraph $H = (V, E)$ whose vertices are the objects we want to color, and whose edges are the structures we want to find.

Proof. Fix the number of colors r and the dimension k. We are coloring k-sets of the natural numbers, and we want to find an n-set such that all of its k-sets have the same color. Define

$$V \coloneqq [\omega]^k, \qquad E \coloneqq \{[A]^k \mid A \in [\omega]^n\}.$$

Since $\left|[\omega]^k\right| = \omega$, assuming the axiom of countable choice AC_ω is sufficient for the compactness principle to hold.

The plan is to show the following:

$$\mathsf{IRT} \implies \chi(H) > r \implies (\exists \text{ finite } W \subseteq V)\, \chi(H_W) > r \implies \mathsf{FRT},$$

where IRT and FRT denote the infinite and finite versions of Ramsey's theorem, respectively.

The infinite version of Ramsey's theorem implies that for every r-coloring of $[\omega]^k$, there exists an *infinite* $A \subseteq \omega$ such that $[A]^k$ is monochromatic. Let A' be the set containing the first n elements of A. Then $[A']^k$ is a monochromatic edge of H. Notice that this implies $\chi(H) > r$, since every r-coloring of H contains a monochromatic edge.

By the compactness principle, there exists a finite $W \subseteq V$ such that $\chi(H_W) > r$: every r-coloring of H_W contains a monochromatic edge. Let $N \in \omega$ be the first number larger than all the natural numbers explicitly mentioned in W. This means that $W \subseteq [N]^k$. Therefore, for every r-coloring of $[N]^k \subseteq V$, there exists $A \subseteq N$ of size n such that $[A]^k$ is monochromatic. We have proved the finite version of Ramsey's theorem. □

We have derived the finite version of Ramsey's theorem from the infinite one, and in doing so we had to assume a weak form of the axiom of choice. However, as one might suspect, the axiom of choice is not required to prove finite Ramsey's theorem.

Proof sketch of the Finite Ramsey Theorem. There are more ways to prove the finite version of Ramsey's theorem, but after examining the first proof of the infinite version presented above, it should be clear that the same method can be used to prove the finite version. In the finite case, we do not need infinitely many t_i to use pigeonhole at the end — $(n-1)r+1$ is surely enough. The only difference is that at each step t_i, we need to invoke the induction hypothesis for a sufficiently large n_i so that each A_i is large enough to keep the induction going. □

Exercise 2.6. Show that for all $n, r, k < \omega$ there exists $N < \omega$ such that $N \longrightarrow (n)^{<k}_r$, meaning that for every mapping $\chi\colon [N]^{<k} \to r$, there exists a subset $A \subseteq N$ of size n such that $f \restriction [A]^m$ is constant for every $m < k$. Note that the value of f on $[A]^m$ might differ for different m.

Exercise 2.7. Show that this statement would be false if we demanded that the value of f on $[A]^m$ should be the same for all $m < k$.

2.3 Paris–Harrington Theorem

Exercise 2.8. Prove, using the compactness principle, the following extension of Ramsey's theorem: For all n, r, $k < \omega$ there exists $N < \omega$ such that for every coloring $\chi\colon [N]^k \to r$ there exists a subset $A \subseteq N$ of size $|A| \geq \min(A)$ and $|A| \geq n$ that is homogeneous for χ.

Theorem 2.9 (Paris–Harrington [PH77])**.** *The above statement cannot be proved in Peano Arithmetic.*

Intuition. While it is not necessary to use compactness to derive finite Ramsey's theorem, it is somehow necessary to use it for this extension.

2.4 Product Ramsey Theorem

For natural numbers $N_1, N_2, n_1, n_2, k_1, k_2$ and r, the expression

$$N_1, N_2 \longrightarrow (n_1, n_2)^{k_1,k_2}_r$$

means that for every r-coloring $\chi\colon [N_1]^{k_1} \times [N_2]^{k_2} \to r$ of tuples of k_1-sets of N_1 and k_2-sets of N_2, there exist subsets $A_1 \subseteq N_1$ and $A_2 \subseteq N_2$ of sizes $|A_1| = n_1$ and $|A_2| = n_2$ such that $[A_1]^{k_1} \times [A_2]^{k_2}$ is monochromatic.

Example 1. $N, N \to (n, n)^{1,1}_2$ means that if we 2-color the edges of the complete bipartite graph $K_{N,N}$, we will find a monochromatic copy of $K_{n,n}$.

Theorem 2.10. *For every number of colors r, dimensions k_1, k_2, and sizes n_1, n_2, there exist numbers N_1, N_2 such that $N_1, N_2 \longrightarrow (n_1, n_2)^{k_1,k_2}_r$.*

Proof. We first focus on the case $N_1, N_2 \longrightarrow (n_1, n_2)^{1,1}_r$ which corresponds to r-coloring the edges of the complete bipartite graph K_{N_1,N_2}. Fix a coloring $\chi\colon N_1 \times N_2 \to r$.

We again use the trick with signatures. It would be nice if we could find a subset $A_2 \subseteq N_2$ of size $|A_2| = n_2$ with the property that for every $v \in N_1$, the color $\chi(v, a)$ is the same for all $a \in A_2$. We denote this color by $c(v)$. In other words, the color signature

$$\tau_a \coloneqq (\chi(0, a), \chi(1, a), \ldots, \chi(N_1 - 1, a))$$

is the same for all $a \in A_2$.

Once we have this set A, we let $N_1 \longrightarrow (n_1)^1_r$. Then there is $A_1 \subseteq N_1$ of size n_1 such that the color $c(v)$ is the same for all $v \in A_1$. Hence $A_1 \times A_2$ is monochromatic.

How do we get this magical subset $A_2 \subseteq N_2$? We treat the signatures τ defined above as new colors of $u \in N_2$. Since there are only r^{N_1} possible signatures, when we let $N_2 \to (n_2)^1_{r^{N_1}}$, we find $A_2 \subseteq N_2$ of size n_2 such that all $a \in A_2$ have the same signature, which is exactly what we wanted.

To show the general arrow $N_1, N_2 \longrightarrow (n_1, n_2)^{k_1,k_2}_r$, use

$$N_1 \longrightarrow (n_1)^{k_1}_r \quad \text{and} \quad N_2 \longrightarrow (n_2)^{k_2}_{r^T} \quad \text{where} \quad T = \binom{N_1}{k_1}.$$

The same argument applies, just now $a \in [N_2]^{k_2}$ are k_2-sets and their signature is a list of $\binom{N_1}{k_1}$ colors, as the elements $v \in [N_1]^{k_1}$ are k_1-sets of N_1. □

Definition 2.11. The expression

$$N_1, N_2, \dots, N_d \longrightarrow (n_1, n_2, \dots, n_d)^{k_1,k_2,\dots,k_d}_r$$

means that for every r-coloring $\chi\colon [N_1]^{k_1} \times [N_2]^{k_2} \times \cdots \times [N_d]^{k_d} \to r$ of d-tuples of k_i-sets of N_i, there exist subsets $A_i \subseteq N_i$ with $|A_i| = n_i$ such that $[A_1]^{k_1} \times [A_2]^{k_2} \times \cdots \times [A_d]^{k_d}$ is monochromatic.

Theorem 2.12. *For every number of colors r, product size d, dimensions $k_1, \dots, k_d$, and sizes $n_1, \dots, n_d$, there exist numbers $N_1, \dots, N_d$ such that $N_1, N_2, \dots, N_d \longrightarrow (n_1, n_2, \dots, n_d)^{k_1,k_2,\dots,k_d}_r$.*

Proof. By induction on d. It is very similar to the case $d = 2$, just take

$$N_1 \longrightarrow (n_1)^{k_1}_r \quad \text{and} \quad N_2, \dots, N_d \longrightarrow (n_2, \dots, n_d)^{k_2,\dots,k_d}_{r^T} \quad \text{where} \quad T = \binom{N_1}{k_1}.$$

This allows us to pick a subset $A \subseteq [N_2]^{k_2} \times \cdots \times [N_d]^{k_d}$ with the proper n_i sizes such that the signature of each $a \in A$ is the same. □

Infinite product Ramsey theorem? We show that the product Ramsey theorem does not extend to infinite products. More precisely, it is *not* true that for every 2-coloring of the infinite complete bipartite graph $K_{\omega,\omega}$, there is a monochromatic copy of $K_{\omega,\omega}$.

Proposition 2.13. $\omega, \omega \not\longrightarrow (\omega, \omega)^{1,1}_2$.

Proof. We define a bad coloring $\chi\colon \omega \times \omega \to \{\text{red}, \text{blue}\}$ by

$$\chi(n, m) \coloneqq \begin{cases} \text{red}, & \text{if } n < m, \\ \text{blue}, & \text{if } n \geq m. \end{cases}$$

Given infinite $N, M \subseteq \omega$, let

$$\begin{aligned} n_0 &\coloneqq \min(N), \\ m_0 &\coloneqq \min\{m \in M \mid m > n_0\}, \\ n_1 &\coloneqq \min\{n \in N \mid n \geq m_0\}. \end{aligned}$$

Then $\chi(n_0, m_0) = \text{red}$ but $\chi(n_1, m_0) = \text{blue}$, so $N \times M$ is not monochromatic. □

3 Halpern–Läuchli Theorem

4 Milliken's Tree Theorem

5 Hindman's Theorem

Hindman's theorem is historically significant because it introduced idempotent ultrafilters to Ramsey theory. We will, however, not explore this direction.

5.1 Results leading up to Hindman's theorem

We denote by $\mathbb{N} \coloneqq \omega \setminus \{0\}$ the set of all positive integers.

Theorem 5.1 (Schur [Sch16])**.** *If $\mathbb{N}$ is finitely colored, there exist integers x, y, and z of the same color such that $x + y = z$.*

Proof. Given a coloring χ of $\mathbb{N}$, define a coloring χ' of $[\mathbb{N}]^2$ by $\chi'(\{a, b\}) \coloneqq \chi(|a - b|)$. By Ramsey's theorem, there are integers $a < b < c$ such that $[\{a, b, c\}]^2$ is monochromatic, so $\chi(b - a) = \chi(c - b) = \chi(c - a)$. Put $x \coloneqq b - a$ and $y \coloneqq c - b$. □

Remark. Schur derived this result 13 years before Ramsey's theorem, and he did it in order to disprove Fermat's last theorem modulo a prime [Sch16]. He showed that for every m and for every sufficiently large prime p, there are numbers $x, y, z \in \mathbb{F}_p$ such that $x^m + y^m \equiv z^m \pmod{p}$.

One way to generalize Schur's theorem is to consider more complex constraints such as $x^2 - y = z$ or $x + y + z = w$. This problem was studied by Rado in his PhD thesis [Rad33]. We state one of his main results.

Theorem 5.2 (Rado, 1933)**.** *Let $S(x_1, \dots, x_n)$ be a linear homogeneous constraint*

$$c_1x_1 + c_2x_2 + \cdots + c_nx_n = 0, \quad c_i \in \mathbb{Z}.$$

Then the two following conditions are equivalent.

(I) *For every finite coloring of $\mathbb{N}$, there are integers $a_1, a_2, \dots, a_n$ of the same color such that $S(a_1, a_2, \dots, a_n)$ holds.*

(II) *Some nonempty subset of the coefficients c_i sums up to zero.*

Remark. The original proof of Rado's theorem relies on a strengthening of van der Waerden's Theorem. Di Nasso and Baglini [NB26] recently found an elementary proof of Rado's theorem.

Another way to generalize Schur's theorem is to demand that some numbers x, y, z have the same color, and all their sums $x + y$, $x + z$, $y + z$, and $x + y + z$ also share that color.

For $X \subseteq \mathbb{N}$, define the set of all finite sums of X to be the set

$$\mathrm{FS}(X) \coloneqq \left\{ \textstyle\sum_{x \in A} x \,\middle|\, \varnothing \neq A \subseteq X \text{ finite} \right\}.$$

In the late 1960s, Jon Folkman, Richard Rado, and Jon Henry Sanders independently extended Schur's theorem in the following way:

Theorem 5.3 (Folkman–Rado–Sanders, c. 1968)**.** *If $\mathbb{N}$ is finitely colored, there exist arbitrarily large finite sets $X \subseteq \mathbb{N}$ such that $\mathrm{FS}(X)$ is monochromatic.*

Hindman [Hin74] extended this even further in 1974. Soon after, Baumgartner [Bau74] found a shorter combinatorial proof of Hindman's theorem on which the proof of Hindman's theorem presented here is based.

Theorem 5.4 (Hindman, 1974)**.** *If $\mathbb{N}$ is finitely colored, there exists an infinite set $X \subseteq \mathbb{N}$ such that $\mathrm{FS}(X)$ is monochromatic.*

5.2 Proof of Hindman's theorem

We denote by $\mathrm{Fin} \subseteq \mathcal{P}(\omega)$ the set of all finite nonempty subsets of natural numbers. A *block scheme* is a family $\mathcal{B} = \langle B_i \,|\, i \in \omega \rangle$ where $B_i \in \mathrm{Fin}$, and if $i < j$, then $\max B_i < \min B_j$. Such a block scheme will sometimes be written as

$$\mathcal{B} = B_0 \oplus B_1 \oplus B_2 \oplus \cdots .$$

Similarly, if we write $X = B_0 \oplus B_1 \oplus \cdots \oplus B_n$, it denotes that $\max B_i < \min B_j$ whenever $i < j \leq n$, and that $X = \{B_i \,|\, i \leq n\}$. If n is an integer, we denote by $\mathcal{B} \restriction n$ the sub-scheme consisting of blocks $B \in \mathcal{B}$ such that $\min B > n$. For a set $D \subseteq \mathrm{Fin}$, define the set of its finite unions as

$$\mathrm{FU}(D) \coloneqq \Big\{ \bigcup A \,\Big|\, \varnothing \neq A \subseteq D \text{ finite} \Big\}.$$

Theorem 5.5. *If Fin is finitely colored, there exists a block scheme $\mathcal{B}$ such that $\mathrm{FU}(\mathcal{B})$ is monochromatic.*

Observe that Theorem 5.5 implies Theorem 5.4. Indeed, given a coloring χ of $\mathbb{N}$, define an injection $j \colon \mathrm{Fin} \to \mathbb{N}$ by $j(A) \coloneqq \sum_{n \in A} 2^n$, and a coloring χ' of Fin by $\chi'(A) \coloneqq \chi(j(A))$. Note that if A and B are disjoint, then $j(A) + j(B) = j(A \cup B)$. Hence, if $\mathrm{FU}(\mathcal{B})$ is monochromatic for a block scheme $\mathcal{B} = \langle B_i \,|\, i \in \omega \rangle$, then $FS(X)$ is monochromatic for the set $X = \{j(B_i) \,|\, i \in \omega\}$.

Definition 5.6. We define a relation $\sqsubseteq$ on block schemes by letting $\mathcal{B}' \sqsubseteq \mathcal{B}$ if $\mathcal{B}' \subseteq \mathrm{FU}(\mathcal{B})$. We say that $S \subseteq \mathrm{Fin}$ is *large* for $\mathcal{B}$ if for every $\mathcal{B}' \sqsubseteq \mathcal{B}$ we have $S \cap \mathrm{FU}(\mathcal{B}') \neq \varnothing$.

Before we continue, observe the following:

1. $\mathcal{B}' \sqsubseteq \mathcal{B} \implies \mathrm{FU}(\mathcal{B}') \subseteq \mathrm{FU}(\mathcal{B})$,
2. $\sqsubseteq$ is a partial order,
3. $\mathrm{FU}(\mathcal{B})$ is large for $\mathcal{B}$. In particular, Fin is large for $[\omega]^1$,
4. if S is large for $\mathcal{B}$ and $\mathcal{B}' \sqsubseteq \mathcal{B}$, then S is large for $\mathcal{B}'$,
5. S is large for $\mathcal{B} \iff S \cap \mathrm{FU}(\mathcal{B})$ is large for $\mathcal{B}$.

Observation 5.7. *If S is large for $\mathcal{B}$ and $S = S_1 \cup S_2 \cup \cdots \cup S_n$, then there is $i \in \{1, 2, \ldots, n\}$ and a block scheme $\mathcal{B}' \sqsubseteq \mathcal{B}$ such that S_i is large for $\mathcal{B}'$.*

Proof. Let $S = S_1 \cup S_2$; the general case follows by induction. If S_1 is not large for $\mathcal{B}$, then there is $\mathcal{B}' \sqsubseteq \mathcal{B}$ such that $S_1 \cap \mathrm{FU}(\mathcal{B}') = \varnothing$. If S_2 were not large for $\mathcal{B}'$, there would be $\mathcal{B}'' \sqsubseteq \mathcal{B}'$ such that $S_2 \cap \mathrm{FU}(\mathcal{B}'') = \varnothing$. But then $S \cap \mathrm{FU}(\mathcal{B}'') = \varnothing$, contradicting that S is large for $\mathcal{B}$. □

Lemma 5.8. *If S is large for $\mathcal{B} = \langle B_i \,|\, i \in \omega \rangle$, then there are $x, y \in \mathrm{FU}(\mathcal{B})$ such that $y, x \cup y \in S$ and $\max x < \min y$.*

Proof. Put $D \coloneqq S \cap \mathrm{FU}(\mathcal{B})$ and observe that if $y \in D$, then there is $z \in D$ such that $\max y < \min z$. Indeed, suppose that there were $y \in D$ such that for all $z \in D$ we had $\min z \leq \max y$. Then the sub-scheme $\mathcal{B}' \coloneqq \mathcal{B} \restriction \max y$ would contradict S being large for $\mathcal{B}$, since $S \cap \mathrm{FU}(\mathcal{B}') = \varnothing$.

Using this, we can construct a block scheme

$$\mathcal{Y} = y_0 \oplus y_0' \oplus y_1 \oplus y_1' \oplus y_2 \oplus y_2' \oplus \cdots \sqsubseteq \mathcal{B},$$

where $y_i, y_i' \in S$ for all $i \in \omega$. Consider the block scheme

$$\mathcal{Y}' \coloneqq (y_0 \cup y_0') \oplus (y_1 \cup y_1') \oplus (y_2 \cup y_2') \oplus \cdots \sqsubseteq \mathcal{B}.$$

Since S is large for $\mathcal{B}$, there is $A \in \mathrm{Fin}$ such that $\bigcup_{i \in A} (y_i \cup y_i') \in S$. Suppose that $A = \{i_0 < i_1 < \cdots < i_n < r\}$ and put $x \coloneqq y_r \cup \bigcup_{j \leq n} (y_{i_j} \cup y_{i_j}')$ and $y \coloneqq y_r'$. □

Lemma 5.9. *If S is large for $\mathcal{B}$, then there exists $x \in \mathrm{FU}(\mathcal{B})$ such that the set*

$$S(x) \coloneqq \{y \in \mathrm{FU}(\mathcal{B}) \mid y, x \cup y \in S \text{ and } \max x < \min y\}$$

is large for some $\mathcal{B}' \sqsubseteq \mathcal{B}$.

Proof. Since Fin is countable, we can enumerate $\mathrm{FU}(\mathcal{B})$ as $\langle x_i \mid i \in \omega \rangle$ such that whenever $i < j$, then $\max x_i \leq \max x_j$. Assume for contradiction that the claim is false and construct a sequence of block schemes $\mathcal{B} \sqsupseteq \mathcal{B}_0 \sqsupseteq \mathcal{B}_1 \sqsupseteq \mathcal{B}_2 \sqsupseteq \cdots$ such that for every $i \in \omega$, the following holds:

(i) $\mathrm{FU}(\mathcal{B}_i) \cap S(x_i) = \varnothing$, and

(ii) if $z \in \mathcal{B}_i$ and $\min z \leq \max x_{i+1}$, then $z \in \mathcal{B}_{i+1}$.

Suppose that such a sequence exists. Property (ii) ensures that it has a limit $\mathcal{B}^* \sqsubseteq \mathcal{B}$ consisting of blocks B such that $B \in \mathcal{B}_i$ for all sufficiently large i. But since $\mathrm{FU}(\mathcal{B}^*) \cap S(x_i) = \varnothing$ for all $i \in \omega$ due to property (i), the existence of this block scheme contradicts Lemma 5.8.

We construct the sequence as follows. Since $S(x_0)$ is not large for $\mathcal{B}$, there is $\mathcal{B}_0 \sqsubseteq \mathcal{B}$ such that $\mathrm{FU}(\mathcal{B}_0) \cap S(x_0) = \varnothing$. If $\mathcal{B}_i$ has already been defined, to define $\mathcal{B}_{i+1}$, we first let $\mathcal{B}_i^- \coloneqq \mathcal{B}_i \restriction \max x_{i+1}$. Denote the blocks we deleted by $z_0, z_1, \ldots, z_n$. Since $S(x_{i+1})$ is not large for $\mathcal{B}_i^-$, there is $\mathcal{B}'_{i+1} \sqsubseteq \mathcal{B}_i^-$ such that $\mathrm{FU}(\mathcal{B}'_{i+1}) \cap S(x_{i+1}) = \varnothing$. Define $\mathcal{B}_{i+1}$ as the block scheme obtained from $\mathcal{B}'_{i+1}$ by adding the blocks $z_0, z_1, \ldots, z_n$ back. Note that still $\mathrm{FU}(\mathcal{B}_{i+1}) \cap S(x_{i+1}) = \varnothing$ because $\min z_j \leq \max x_{i+1}$ for all $j \leq n$. □

Lemma 5.10. *If S is large for $\mathcal{B}$, then there is $\mathcal{B}^* \sqsubseteq \mathcal{B}$ such that $\mathrm{FU}(\mathcal{B}^*) \subseteq S$.*

Proof. First, construct a block scheme $\mathcal{X} = x_0 \oplus x_1 \oplus x_2 \oplus \cdots \sqsubseteq \mathcal{B}$ by repeated application of Lemma 5.9. We do this by building sequences x_n, $\mathcal{B}_n$, and S_n where

(1) $\mathcal{B}_0 = \mathcal{B}$ and $S_0 = S$,

(2) $\mathcal{B}_{n+1} \sqsubseteq \mathcal{B}_n$ and $S_{n+1} \subseteq S_n \cap \mathrm{FU}(\mathcal{B}_n)$,

(3) S_n is large for $\mathcal{B}_n$,

(4) $x_n \in \mathrm{FU}(\mathcal{B}_n)$, and if $y \in S_{n+1}$, then $x_n \cup y \in S_n$ and $\max x_n < \min y$.

By combining properties (2) and (4), we also get

(5) if $y \in S_m$ where $m > n$, then $x_n \cup y \in S_n$.

Next, construct a block scheme

$$\mathcal{B}^* = B_0 \oplus B_1 \oplus B_2 \oplus \cdots \sqsubseteq \mathcal{X}.$$

Choose $B_0 \in S \cap \mathrm{FU}(\mathcal{X})$ arbitrarily (this is possible since S is large for $\mathcal{B}$), and for $n > 1$ select $B_n \in S_{k_n+1} \cap \mathrm{FU}(\mathcal{X})$ where $k_n \coloneqq \max\{k \mid x_k \subseteq B_{n-1}\}$. (The blocks B_j are made up of smaller pieces $x - i$. The index k_n is the index of the last piece in B_{n-1}.) This is possible because $\mathcal{X} \restriction \max x_{k_n} \sqsubseteq \mathcal{B}_{k_n+1}$, and S_{k_n+1} is large for $\mathcal{B}_{k_n+1}$. Note that $\max B_{n-1} < \min B_n$ since $\max B_{n-1} = \max x_{k_n}$.

We claim that $\mathcal{B}^*$ has the desired property. Let $A \in \mathrm{FU}(\mathcal{B}^*)$ and suppose that

$$A = B_{i_1} \oplus B_{i_2} \oplus \cdots \oplus B_{i_n} \oplus B_r.$$

for some indices $i_1 < i_2 < \cdots < i_n < r$. Since $\mathcal{B}^* \sqsubseteq \mathcal{X}$, we can write

$$B_{i_1} \oplus B_{i_2} \oplus \cdots \oplus B_{i_n} = x_{j_1} \oplus x_{j_2} \oplus \cdots \oplus x_{j_m}$$

for some indices $j_1 < j_2 < \cdots < j_m$. Note that $j_m \leq k_r$. Since $B_r \in S_{k_r+1} \subseteq S_{j_m+1}$, we have $x_{j_m} \cup B_r \in S_{j_m}$ by property (5). Thus $x_{j_{m-1}} \cup x_{j_m} \cup B_r \in S_{j_{m-1}}$ again by property (5) because $j_m > j_{m-1}$. By repeating this, we see that

$$A = x_{j_1} \cup \cdots \cup x_{j_{m-1}} \cup x_{j_m} \cup B_r \in S_{j_1} \subseteq S.$$ □

We can now finish the proof of Hindman's theorem.

Proof of Theorem 5.5. Let Fin be r-colored; that is, partitioned into r classes

$$\mathrm{Fin} = S_1 \cup S_2 \cup \cdots \cup S_r.$$

Since Fin is large for $[\omega]^1$, by Observation 5.7 there is a color i and a block scheme $\mathcal{B} \sqsubseteq [\omega]^1$ such that S_i is large for $\mathcal{B}$. By Lemma 5.10, there is $\mathcal{B}^* \sqsubseteq \mathcal{B}$ such that $\mathrm{FU}(\mathcal{B}^*) \subseteq S_i$. □

Remarks about the proof Hindman's theorem implies all of the lemmas we used to prove it. Therefore, if one believes that Hindman's theorem is true, one must trust these lemmas even before proving them.

To demonstrate this, we will show that Theorem 5.5 implies Lemma 5.10. Suppose that S is large for $\mathcal{B} = \langle B_i \,|\, i \in \omega \rangle$. Color $A \in \mathrm{Fin}$ red if $g(A) \coloneqq \bigcup_{i \in A} B_i \in S$, and blue otherwise. By Theorem 5.5, there is a homogeneous block scheme

$$\mathcal{X} = \langle x_n \,|\, n \in \omega \rangle \sqsubseteq [\omega]^1.$$

It defines a block scheme $\mathcal{B}^* = \langle D_n \,|\, n \in \omega \rangle \sqsubseteq \mathcal{B}$ where $D_n = g(x_n)$. If $\mathrm{FU}(\mathcal{X})$ is red, then $\mathrm{FU}(\mathcal{B}^*) \subseteq S$, and we are happy. If $\mathrm{FU}(\mathcal{X})$ is blue, then $\mathrm{FU}(\mathcal{B}^*) \cap S = \varnothing$, contradicting the fact that S is large for $\mathcal{B}$.

5.3 Product version of Hindman's theorem

Similar to finite sums, we may consider finite products

$$\mathrm{FP}(X) \coloneqq \Big\{ \textstyle\prod_{x \in A} x \,\Big|\, \varnothing \neq A \subseteq X \text{ finite} \Big\}.$$

of a set $X \subseteq \mathbb{N}$. Observe that Hindman's theorem implies the following.

Corollary 5.11. *If $\mathbb{N}$ is finitely colored, there exists an infinite set $X \subseteq \mathbb{N}$ such that $\mathrm{FP}(X)$ is monochromatic.*

Proof. Given a finite coloring $\chi \colon \mathbb{N} \to r$, define a new coloring χ' by $n \mapsto \chi(2^n)$. Hindman's theorem for χ' yields an infinite set X. For finite $A \subseteq X$ we then have

$$\chi'\big(\sum_{x \in A} x\big) = \chi\big(2^{\sum_{x \in A} x}\big) = \chi\big(\prod_{x \in A} 2^x\big),$$

so the set $\{2^x \,|\, x \in X\}$ works for the product version in χ. □

It is natural to ask whether it is possible to combine this: that given any finite coloring of $\mathbb{N}$ we can find an infinite $X \subseteq \mathbb{N}$ such that both FS(X) and $\mathrm{FP}(X)$ are monochromatic with the same color. Unfortunately, this theorem is false [Hin80]. However, in their 2026 math dump, OpenAI proved the following theorem.

Theorem 5.12 (OpenAI, 2026)**.** *If $\mathbb{N}$ is finitely colored, there exist arbitrarily large finite sets $X \subseteq \mathbb{N}$ such that* FS(X) *and* FP(X) *are monochromatic with the same color.*

6 Hales–Jewett Theorem

The symbol Σ will always denote a finite set called the *alphabet*. Given $n \leq \omega$, we denote by Σ^n the set of all words over Σ of length n (formally functions $f\colon n \to \Sigma$), by $\Sigma^{\leq n}$ the set of all words over Σ of length at most n, and by $\Sigma^{<n}$ the set $\Sigma^{\leq n} \setminus \Sigma^n$. Given a word U, we denote by $|U|$ its length. For every $i < |U|$, we denote by U_i the letter of U at index i. Indices start from 0. Given words U and V, we say that U *extends* V if $|U| > |V|$ and for every $i < |V|$ it holds that $U_i = V_i$. Given a finite word U and another word V, we denote their concatenation by $U^\frown V$. Formally this word extends U and for $j \geq |U|$ we have $(U^\frown V)_j = V_{j-|U|}$. We denote by $\varnothing$ the empty word of length 0.

The symbols λ_i for $i \in \omega$ are called *variables* and we assume that none of them appears in the alphabet. An *n-variable word* over Σ is a word W over the alphabet $\Sigma \cup \{\lambda_i : i < n\}$ such that for every $i < n$, the variable λ_i occurs in W, and if $i > 0$, then the last occurrence of λ_{i-1} is before the first occurrence of λ_i. An *n-parameter word* over Σ is defined analogously, but this time we demand that the first occurrence of λ_i always appears before the first occurrence of λ_{i+1}. By $\mathrm{Lines}(\Sigma)$, we denote the set of all 1-variable words of finite length, and we call them *(combinatorial) lines*.

Let W be an n-variable word, and let U be a word of length k where $0 \leq k \leq n \leq \omega$. Then we denote by $W(U)$ the word created by *substituting* U into W. More precisely, $W(U)$ is the word obtained from W by replacing each occurrence of λ_i with U_i for $0 \leq i < k$ and truncating the result just before the first occurrence of λ_k in W (if $n > k$). Note that if U is an m-variable word, then $W(U)$ is also an m-variable word satisfying $W(U(v)) = W(U)(v)$ for all $v \in \Sigma^{\leq m}$. For example:

$$W = 0\lambda_0\lambda_1 2\lambda_1\lambda_2\lambda_3 23\lambda_4 0; \quad W(\varnothing) = 0; \quad U = \lambda_0 1\lambda_0\lambda_1; \quad W(U) = 0\lambda_0 121\lambda_0\lambda_1 23;$$
$$v = 47; \quad U(v) = 4147; \quad W(4147) = W(U)(47) = 041214723.$$

Finally, if W is an n-variable word and S is a set of words of length at most n, then we denote by $W[S]$ the set $\{W(u) : u \in S\}$.

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Definition 6.1. $\mathrm{HJ}(k, d, r)$ is the least $N \in \omega$ such that for every alphabet Σ of size $|\Sigma| = k$ and every coloring $\chi\colon \Sigma^N \to r$, there exists a d-parameter word W over Σ of length N such that $W[\Sigma^d]$ is monochromatic.

Proof scheme:

1. $\mathrm{HJ}(2, 1, \forall r)$ and $\mathrm{HJ}(\forall k, 1, 1)$
2. $\mathrm{HJ}(k, d, \forall r) \implies \mathrm{HJ}(k, d+1, \forall r)$
3. $\mathrm{HJ}(k, \forall d, \forall r) \wedge \mathrm{HJ}(k+1, 1, r-1) \implies \mathrm{HJ}(k+1, 1, r)$

Proof of 1 easy, proof of 2 is fusion, and proof of 3 as follows.

Increasing the size of the alphabet. Suppose $\Pi = \Sigma \cup \{\alpha\}$ with $|\Sigma| = k$ and $\alpha \notin \Sigma$. Fix $r \geq 2$. Put $h := \mathrm{HJ}(k+1, 1, r-1)$ and $d := h + 1$ and $N := \mathrm{HJ}(k, d, r)$. Given $\chi\colon \Pi^N \to r$ let χ' be its restriction to Σ^n. By the choice of N there is a d-parameter word W over Σ such that $W[\Sigma^d]$ is monochromatic with color c.

Consider the two following cases:

(a) There is $u \in \Pi^d$ containing α with $\chi(W(u)) = c$. Then let L be the combinatorial line obtained from $W(u)$ by replacing each occurrence of α with the variable λ_0. Observe that $L[\Pi]$ is monochromatic with color c. Indeed, $L(\alpha) = W(u)$ and for $\beta \in \Sigma$ we have $L(\beta) \in W[\Sigma^d]$.

(b) Otherwise for all $u \in \Pi^d$ containing α we have $\chi(W(u)) \neq c$. This means that when we substitute α for λ_0 in W, we will never get color c. Consider the $(d-1)$-dimensional subspace $U := W(\alpha, \lambda_0, \dots, \lambda_{d-2})$. It is isomorphic to the $(d-1)$-dimensional space $\lambda_0\lambda_1 \dots \lambda_{d-2}$. Define a coloring $\chi' \colon \Pi^{d-1} \to r$ by $\chi'(u) = \chi(U(u))$, noting that $c \notin \mathrm{Rng}(\chi')$. We have just defined a coloring of a $d-1 = h$-dimensional space over Π using $r-1$ colors, so by our choice of h there exists a combinatorial line L such that $L[\Pi^{d-1}]$ is χ'-monochromatic. Then $U(L)$ is a χ-monochromatic combinatorial line of length N. □.

7 Graham–Rothschild Theorem

8 Carlson–Simpson Lemma

The Carlson–Simpson lemma, appearing as Theorem 6.3 in [CS84] is an infinite-dimensional version of the Hales–Jewett theorem. We will first prove this much mode powerful statement and then derive the Hales–Jewett theorem as a corollary.

The proof we present is due to Jan Hubička (unpublished). It is inspired by Baumgartner's proof of Hindman's Theorem [Bau74] and remains close in spirit to the original argument of Carlson and Simpson [CS84, Theorem 6.3]. The main distinction is that this approach does not require the Hales–Jewett theorem as a black box.

Combinatorial forcing The astute reader will notice that the high-level strategy of the proof is very similar to the proof of Hindman's theorem presented in Section 5. This method is called the method of *combinatorial forcing*.

Notation The symbol Σ will always denote a finite set called the *alphabet*. Given $n \leq \omega$, we denote by Σ^n the set of all words over Σ of length n (formally functions $f \colon n \to \Sigma$), by $\Sigma^{\leq n}$ the set of all words over Σ of length at most n, and by $\Sigma^{<n}$ the set $\Sigma^{\leq n} \setminus \Sigma^n$. Given a word U, we denote by $|U|$ its length. For every $i < |U|$, we denote by U_i the letter of U at index i. Indices start from 0. Given words U and V, we say that U *extends* V if $|U| > |V|$ and for every $i < |V|$ it holds that $U_i = V_i$. We denote by $\varnothing$ the empty word of length 0.

The symbols λ_i for $i \in \omega$ are called *variables* and we assume that none of them appears in the alphabet. An *n-variable word* over Σ is a word W over the alphabet $\Sigma \cup \{\lambda_i : i < n\}$ such that for every $i < n$, the variable λ_i occurs in W, and if $i > 0$, then the last occurrence of λ_{i-1} is before the first occurrence of λ_i. By $\mathrm{Lines}(\Sigma)$, we denote the set of all 1-variable words of finite length, and we call them *lines*. By $\mathrm{Var}(\Sigma)$, we denote the set of all ω-variable words and call them *subspaces*.

Let W be an n-variable word, and let U be a word of length k where $0 \leq k \leq n \leq \omega$. Then we denote by $W(U)$ the word created by *substituting* U into W. More precisely, $W(U)$ is the word obtained from W by replacing each occurrence of λ_i with U_i for $0 \leq i < k$ and truncating the result just before the first occurrence of λ_k in W (if $n > k$). Note that if U is an m-variable word, then $W(U)$ is also an m-variable word satisfying $W(U(v)) = W(U)(v)$ for all $v \in \Sigma^{\leq m}$. For example:

$$W = 0\lambda_0\lambda_1 2\lambda_1\lambda_2\lambda_3 23\lambda_4 0; \quad W(\varnothing) = 0; \quad U = \lambda_0 1\lambda_0\lambda_1; \quad W(U) = 0\lambda_0 121\lambda_0\lambda_1 23;$$
$$v = 47; \quad U(v) = 4147; \quad W(4147) = W(U)(47) = 041214723.$$

Finally, if W is an n-variable word and S is a set of words of length at most n, then we denote by $W[S]$ the set $\{W(u) : u \in S\}$.

Theorem 8.1 (1-dimensional $\star$-version of the Hales–Jewett theorem)**.** *For every finite set Σ and every finite coloring χ of $\Sigma^{<\omega}$, there exists $L \in \mathrm{Lines}(\Sigma)$ such that $L[\Sigma^{<2}]$ is monochromatic: there exists a color c such that $\chi(L(v)) = c$ for every $v \in \Sigma^{<2}$.*

Theorem 8.2 (ω-dimensional $\star$-version of the Hales–Jewett theorem)**.** *For every finite set Σ and every finite coloring χ of $\Sigma^{<\omega}$, there exists $W \in \mathrm{Var}(\Sigma)$ such that $W[\Sigma^{<\omega}]$ is monochromatic: there exists a color c such that $\chi(W(v)) = c$ for every $v \in \Sigma^{<\omega}$.*

Note that both of these theorems are true when $|\Sigma| = 1$ from Dirichlet's principle. Also observe that it is enough to prove them for coloring with only 2 colors, since the general case then follows by a simple inductive argument.

From lines to subspaces by combinatorial forcing

In this section, we fix a finite alphabet Σ and assume that Theorem 8.1 is true for this alphabet. Using this assumption, we prove that Theorem 8.2 holds for Σ as well.

Definition 8.3 (Partial order of subspaces)**.** Given $U, W \in \mathrm{Var}(\Sigma)$, we write $U \leq W$ if there exists $V \in \mathrm{Var}(\Sigma)$ such that $U = W(V)$.

Definition 8.4 (Large set)**.** Let $\mathcal{O}$ be a subset of $\Sigma^{<\omega}$. We call $\mathcal{O}$ *large* if for every $W \in \mathrm{Var}(\Sigma)$ there exists $u \in \Sigma^{<\omega}$ such that $W(u) \in \mathcal{O}$.

We say that an n-variable word U *avoids* $\mathcal{O} \subseteq \Sigma^{<\omega}$ if there is no $u \in \Sigma^{\leq n}$ such that $U(u) \in \mathcal{O}$. Hence $\mathcal{O}$ is large if and only if no ω-variable word avoids $\mathcal{O}$.

The main result of this section is the following proposition.

Proposition 8.5. *If $\mathcal{O} \subseteq \Sigma^{<\omega}$ is large then there exists $W \in \mathrm{Var}(\Sigma)$ such that $W[\Sigma^{<\omega}] \subseteq \mathcal{O}$.*

Towards proving Proposition 8.5, we start with two observations that follow directly from the definition of a large set.

Observation 8.6. *If $\mathcal{O} \subseteq \Sigma^{<\omega}$ is large and $W \in \mathrm{Var}(\Sigma)$ then the set $\{v : W(v) \in \mathcal{O}\}$ is large.*

Proof. Let $U \in \mathrm{Var}(\Sigma)$. Since $\mathcal{O}$ is large and $W(U) \in \mathrm{Var}(\Sigma)$, there is u such that $W(U)(u) = W(U(u)) \in \mathcal{O}$. Hence $U(u) \in \{v : W(v) \in \mathcal{O}\}$. □

Observation 8.7. *If $\Sigma^{<\omega} = \mathcal{O}_0 \cup \mathcal{O}_1$ then there exist $W \in \mathrm{Var}(\Sigma)$ and $i \in \{0, 1\}$ such that $\{v : W(v) \in \mathcal{O}_i\}$ is large.*

Proof. Let $W \in \mathrm{Var}(\Sigma)$ be the identity subspace $\lambda_0\lambda_1\lambda_2\ldots$ and so on. If $\{v : W(v) \in \mathcal{O}_0\} = \mathcal{O}_0$ is not large, there exists $U \in \mathrm{Var}(\Sigma)$ such that $U(v) \notin \mathcal{O}_0$ for every $v \in \Sigma^{<\omega}$. But then $\{v : U(v) \in \mathcal{O}_1\} = \{v : U(v) \in \mathcal{O}\}$ is large by the previous observation. □

Given a finite word U and another word V, we denote their concatenation by $U^\frown V$. We now prove the 1-dimensional version of Proposition 8.5.

Lemma 8.8. *If $\mathcal{O} \subseteq \Sigma^{<\omega}$ is large then there exists $L \in \mathrm{Lines}(\Sigma)$ such that $L[\Sigma^{<2}] \subseteq \mathcal{O}$.*

Proof. Because $\mathcal{O}$ is large, there exists $n \in \omega$ and an n-variable word U avoiding $\mathcal{O}$ with the property that there is no $(n+1)$-variable word U' that extends U and also avoids $\mathcal{O}$. (If the converse were true, we would be able to greedily construct an ω-variable word avoiding $\mathcal{O}$, which would contradict the largeness of $\mathcal{O}$.)

Fix such a word U. Define a finite coloring χ of $\Sigma^{<\omega}$ such that the color $\chi_v = \chi(v)$ of each $v \in \Sigma^{<\omega}$ is a function $\chi_v \colon \Sigma^n \to \{0, 1\}$ satisfying that for every $u \in \Sigma^n$ we have

$$\chi_v(u) = 1 \text{ if and only if } U(u)^\frown v \in \mathcal{O}.$$

Let L' be a monochromatic line given by Theorem 8.1 for the coloring χ and let $c \colon \Sigma^n \to \{0, 1\}$ be the color assigned to every $v \in L'[\Sigma^{<2}]$.

First, consider the case that $c(u) = 0$ for every $u \in \Sigma^n$. In this case, we observe that $U^\frown L'(\lambda_n)$ is an $(n+1)$-variable word that extends U and avoids $\mathcal{O}$, contradicting our choice of U. Consequently, there exists $u \in \Sigma^n$ with $c(u) = 1$. From the definition of the coloring χ and our choice of L', we see that $U(u)^\frown L'(w) \in \mathcal{O}$ for every $w \in \Sigma^{<2}$. So if we put $L \coloneqq U(u)^\frown L'$, then $L[\Sigma^{<2}] \subseteq \mathcal{O}$. □

For the following, we will need to split variable words into a prefix and a tail, and have a way to reconstruct the word from its parts. For $n \leq \omega$, we denote by Id^n the *identity* n-variable word of length n satisfying $\mathrm{Id}^n_i = \lambda_i$ for every $i < n$. Given an ω-variable word W and $n \in \omega$, we denote by $\mathrm{Shift}(W, n)$ the ω-variable word created by concatenating Id^n with the word obtained from W by replacing each occurrence of the variable λ_i with λ_{i+n} for all $i \in \omega$.

Observation 8.9. *Let $W, U \in \mathrm{Var}(\Sigma)$ and let $W' = W(\mathrm{Shift}(U, n))$. If $u \in \Sigma^n$ and $v \in \Sigma^{<\omega}$, then $W'(u^\frown v) = W(u^\frown U(v))$.*

Also note that if $W \in \mathrm{Var}(\Sigma)$ is a subspace and $n \in \omega$, then $W(\mathrm{Id}^n)$ is the word obtained by truncating W just before the first occurrence of the variable λ_n.

Lemma 8.10. *If $\mathcal{O}$ is large, then there exists $W \in \mathrm{Var}(\Sigma)$ such that*

(I) *$W(\varnothing) \in \mathcal{O}$, and*

(II) *the set*
$$\mathcal{O}^+ \coloneqq \{v \in \Sigma^{<\omega} : (\forall c \in \Sigma)\, W(c^\frown v) \in \mathcal{O}\}$$
is large.

Remark. We slightly abuse notation here and identify the letter $c \in \Sigma$ with the word $c \in \Sigma^1$.

Proof. Enumerate $\mathrm{Lines}(\Sigma)$ as $L^0, L^1, \ldots$, such that for every $i < j$ it holds that $|L^i| \leq |L^j|$. We start (and eventually fail) constructing a sequence of subspaces $\mathrm{Id}^\omega = W^0 \geq W^1 \geq W^2 \geq \cdots$ such that for all $i \in \omega$ the following is satisfied:

(i) $W^{i+1}(\mathrm{Id}^{|L^i|}) = W^i(\mathrm{Id}^{|L^i|})$, and

(ii) for every $v \in \Sigma^{<\omega}$ and $j \leq i$ it holds that $W^{i+1}(L^{j\frown} v)[\Sigma^{<2}] \not\subseteq \mathcal{O}$.

We proceed by induction on i. Put $W^0 \coloneqq \mathrm{Id}^\omega$.

Now assume that W^i has already been constructed. It satisfies condition (ii) for $j < i$, and we would like W^{i+1} to satisfy it for $j = i$ as well. Let

$$\mathcal{O}^i \coloneqq \{v \in \Sigma^{<\omega} : W^i(L^{i\frown} v)[\Sigma^{<2}] \subseteq \mathcal{O}\}$$

be the set of all v for which W^i fails condition (ii) when $j = i$. Consider two cases:

(a) If $\mathcal{O}^i$ is large, then $W \coloneqq W^i(L^{i\frown}\lambda_1\lambda_2\lambda_3\ldots)$ has the desired property. Part (I) is satisfied since $W^i(L^{i\frown} v)(\varnothing) = W^i(L^i(\varnothing)) = W(\varnothing)$ and $W^i(L^{i\frown} v)(\varnothing) \in \mathcal{O}$ for any $v \in \mathcal{O}^i$, and part (II) holds because the large set $\mathcal{O}^i$ is a subset of the set $\mathcal{O}^+$ for W.

(b) If $\mathcal{O}^i$ is not large, let $U^i \in \mathrm{Var}(\Sigma)$ be a subspace avoiding $\mathcal{O}^i$. Now let $n_i = |L^i|$ and put
$$W^{i+1} \coloneqq W^i(\mathrm{Shift}(U^i, n_i)).$$
It is easy to see that W^{i+1} satisfies condition (i). By Observation 8.9 we have
$$W^{i+1}(L^{i\frown} v) = W^i(L^{i\frown} U^i(v))$$

and hence $W^{i+1}(L^i{}^\frown v)[\Sigma^{<2}] = W^i(L^i{}^\frown U^i(v))[\Sigma^{<2}] \not\subseteq \mathcal{O}$ for every $v \in \Sigma^{<\omega}$ since we chose U^i to avoid $\mathcal{O}^i$. Therefore W^{i+1} satisfies condition (ii) for $j = i$. To see that it also satisfies it for $j < i$, observe that

$$W^{i+1}(L^j{}^\frown v) = W^i\big(L^j{}^\frown \mathrm{Shift}(U^i, n_i - |L^j|)(v)\big)$$

for every $v \in \Sigma^{<\omega}$. From our assumption that W^i satisfies condition (ii) for $j < i$ and the fact that $\mathrm{Shift}(U^i, n_i - |L^i|)(v) \in \Sigma^{<\omega}$, we get that $W^{i+1}(L^j{}^\frown v)[\Sigma^{<2}] \not\subseteq \mathcal{O}$.

Finally, suppose that we succeeded in constructing W^i for all $i \in \omega$. Let $W \in \mathrm{Var}(\Sigma)$ be the limit of this sequence (which exists thanks to condition (i) and the way we enumerated $\mathrm{Lines}(\Sigma)$). By Observation 8.6, we know that $\mathcal{O}' \coloneqq \{v : W(v) \in \mathcal{O}\}$ is large. Let L be given by Lemma 8.8 for $\mathcal{O}'$. It follows that $W(L)[\Sigma^{<2}] \subseteq \mathcal{O}$. Furthermore, if j satisfies $L_j = L$, then by condition (ii) for $v = \varnothing$, we see that $W^i(L)[\Sigma^{<2}] \not\subseteq \mathcal{O}$ for all $i > j$. Thus also $W(L)[\Sigma^{<2}] \not\subseteq \mathcal{O}$, a contradiction. □

Proof of Proposition 8.5. We define a sequence of subspaces $\mathrm{Id}^\omega = W^0 \geq W^1 \geq W^2 \geq \cdots$ and large sets $\mathcal{O} = \mathcal{O}^0, \mathcal{O}^1, \mathcal{O}^2, \ldots$ as follows. To construct W^{i+1} and $\mathcal{O}^{i+1}$ from W^i and $\mathcal{O}^i$, first apply Lemma 8.10 to $\mathcal{O}^i$, yielding a subspace U^i and a set $\mathcal{O}^{i+1}$, and then put

$$W^{i+1} \coloneqq W^i(\mathrm{Shift}(U^i, i)).$$

Notice that for all $i \in \omega$, the initial segments of W^{i+1} and W^i agree at least up to the last occurrence of λ_{i-1}. Consequently, this sequence of subspaces has a limit W. Proposition 8.5 then follows by clause (i) of the following claim:

Claim 8.10.1. *The following two invariants are satisfied for every $i > 0$:*

(i) *For every word $u \in \Sigma^{<i}$ it holds that $W^i(u) \in \mathcal{O}$.*

(ii) *For every pair of words $u \in \Sigma^i$ and $v \in \mathcal{O}^i$ it holds that $W^i(u^\frown v) \in \mathcal{O}$.*

For $i = 0$, invariant (i) is vacuously true and invariant (ii) follows directly since $\mathcal{O}^0 = \mathcal{O}$ and $\mathrm{Id}^\omega(\varnothing^\frown v) = v$. Now assume that the invariants hold for i. We first verify that invariant (i) holds for $i + 1$. Since W^{i+1} and W^i agree on their initial segments at least up to the last occurrence of λ_{i-1}, it is only necessary to show that $W^{i+1}(u) \in \mathcal{O}$ when $u \in \Sigma^i$. So let $u \in \Sigma^i$ and observe that then $W^{i+1}(u) = W^i(u^\frown U^i(\varnothing))$. Since $U^i(\varnothing) \in \mathcal{O}^i$ by our choice of U^i and Lemma 8.10 (I), it follows that $W^{i+1}(u) \in \mathcal{O}$ from invariant (ii) for i.

In order to verify invariant (ii) for $i + 1$, let $u \in \Sigma^{i+1}$ and $v \in \mathcal{O}^{i+1}$. Write $u = w^\frown c$ where $w \in \Sigma^i$ and $c \in \Sigma^1$. Then $W^{i+1}(u^\frown v) = W^i(w^\frown U^i(c^\frown v))$. Since $v \in \mathcal{O}^{i+1}$, by Lemma 8.10 (II) we have that $U^i(c^\frown v) \in \mathcal{O}^i$, and invariant (ii) for i gives $W^{i+1}(u^\frown v) \in \mathcal{O}$. □

Using Proposition 8.5, we now show that if Theorem 8.1 holds for an alphabet Σ, then Theorem 8.2 holds for the same alphabet as well.

Proof that Theorem 8.1 for Σ implies Theorem 8.2 for Σ. Assume that Theorem 8.1 holds for a fixed alphabet Σ. Let χ be a 2-coloring of $\Sigma^{<\omega}$. By Observation 8.7, there exists a subspace W and a color $i \in \{0, 1\}$ such that $\mathcal{O} \coloneqq \{v : \chi(W(v)) = i\}$ is large. Let U be given by Proposition 8.5 for $\mathcal{O}$. It follows that $W(U)$ is monochromatic since $W(U(v)) = W(U)(v)$ for all $v \in \Sigma^{<\omega}$. □

Increasing the size of the alphabet

It remains to show that Theorem 8.1 holds for every finite alphabet Σ.

Proof of Theorem 8.1. We proceed by induction on the size of the alphabet. If $|\Sigma| = 1$, then the theorem trivially follows from the pigeonhole principle.

Assume that $|\Sigma| > 1$ and choose a letter $c \in \Sigma$. Put $\Pi \coloneqq \Sigma \setminus \{c\}$. By our induction hypothesis, Theorem 8.1 holds for Π, and thus Theorem 8.2 does as well. Fix a 2-coloring χ of $\Sigma^{<\omega}$. We proceed in three steps:

1. Let $W_1 \in \mathrm{Var}(\Pi)$ be a monochromatic subspace given by Theorem 8.2 for the coloring $\chi_1 \coloneqq \chi \restriction \Pi^{<\omega}$ of $\Pi^{<\omega}$ and let α_1 be its color (the χ_1-color of every $w \in W_1[\Pi^{<\omega}]$).

2. Let $W_2 \in \mathrm{Var}(\Pi)$ be a monochromatic subspace given by Theorem 8.2 for the coloring χ_2 of $\Pi^{<\omega}$ defined as $\chi_2(u) \coloneqq \chi(W_1(c^\frown u))$ and let α_2 be its color.

3. Let $W_3 \in \mathrm{Var}(\Pi)$ be a monochromatic subspace given by Theorem 8.2 for the coloring χ_3 of $\Pi^{<\omega}$ defined as $\chi_3(u) \coloneqq \chi(W_1(c^\frown W_2(c^\frown u)))$ and let α_3 be its color.

Given a subspace W, denote by W^+ the word obtained from W by replacing each occurrence of the variable λ_i with λ_{i+1} for all $i \in \omega$. Define a subspace

$$U \coloneqq W_1\big(\lambda_0{}^\frown W_2(\lambda_0{}^\frown W_3^+)^+\big) = W_1\big(\mathrm{Shift}\big(W_2(\mathrm{Shift}(W_3, 1)), 1\big)\big).$$

Now consider the three following possible cases:

(a) If $\alpha_1 = \alpha_2$, then $U(\lambda_0)$ is a monochromatic line of color $\alpha_1 = \alpha_2$. To see this, first notice that if $a \in \Pi$, then $U(a) = W_1(v)$ for some suitable $v \in \Pi^{<\omega}$. Therefore $\chi(U(a)) = \chi_1(U(a)) = \alpha_1$ by our choice of W_1. Similarly, $\chi(U(\varnothing)) = \alpha_1$. Lastly, $\chi(U(c)) = \alpha_2$. Indeed, observe that $U(c) = W_1(c^\frown W_2(\varnothing))$, so $\chi(U(c)) = \chi(W_1(c^\frown W_2(\varnothing))) = \chi_2(W_2(\varnothing)) = \alpha_2$.

(b) If $\alpha_1 = \alpha_3$, then $U(\lambda_0\lambda_0)$ is a monochromatic line of color $\alpha_1 = \alpha_3$. We argue as before, the only difference being that $U(cc) = W_1\big(c^\frown W_2(\lambda_0{}^\frown W_3^+)(c)\big) = W_1\big(c^\frown W_2(c^\frown W_3(\varnothing))\big)$. Thus $\chi(U(cc)) = \chi_3(W_3(\varnothing)) = \alpha_3$.

(c) If $\alpha_2 = \alpha_3$, then $U(c\lambda_0)$ is a monochromatic line of color $\alpha_2 = \alpha_3$. We already know that $\chi(U(c)) = \alpha_2$ and $\chi(U(cc)) = \alpha_3$. In addition, for every $a \in \Pi$ we have $U(ca) = W_1\big(c^\frown W_2(a^\frown W_3(\varnothing))\big)$, so $\chi(U(ca)) = \chi_2(W_2(a^\frown W_3(\varnothing))) = \alpha_2$. □

9 Todorčević–Tyros Theorem

Appendix

A.1 Ordinal Numbers

Definition A.1. A partially ordered class $(Q, <)$ is *well-founded* if every nonempty subset $A \subseteq Q$ has a minimal element. A well-founded linear order is called a *well-order*.

Observation A.2. *A partially ordered class Q is well-founded $\iff$ Q admits no infinite decreasing sequences $x_0 > x_1 > x_2 > \cdots$.*

Proof. "$\Rightarrow$" The set of elements forming an infinite decreasing chain has no minimal element.

"$\Leftarrow$" Suppose a nonempty subset $A \subseteq Q$ has no minimal element. Then for every $x \in A$, there is $x' \in A$ such that $x' \leq x$ but $x \not\leq x'$, so $x' < x$. This allows us to construct an infinite decreasing sequence.[3] □

We use *von Neumann ordinals*, meaning that natural numbers are defined as

$$0 \coloneqq \varnothing,\ 1 \coloneqq \{0\},\ 2 \coloneqq \{0, 1\},\ \ldots,\ n+1 \coloneqq \{0, 1, \ldots, n\} = n \cup \{n\}.$$

The *set of all natural numbers* is denoted by ω. *Ordinal numbers* are a way to generalize natural numbers. Intuitively, they represent the *types of well-ordered sets*. Imagine we are trying to label the elements of a well-ordered set: we label the least element by 0, the next by 1, and so on. The *type* of the well-ordering is then the first label we did not have to use. But what if we run out of labels? Consider the following well-ordering of natural numbers:

$$1 \prec 2 \prec 3 \prec 4 \prec 5 \prec \cdots \prec 0$$

If we label the elements from left to right, all natural numbers are used for the segment $1 \prec 2 \prec \cdots$, leaving no label for 0. This is precisely why we need ordinal numbers: we assign the label ω to 0, and the type of this ordering is $\omega + 1$. Similarly, the well-ordering

$$0 \prec 2 \prec 4 \prec 6 \prec \cdots \prec 1 \prec 3 \prec 5 \prec 7 \prec \cdots$$

has type $\omega + \omega$, because we use all naturals n to label the even numbers, and then all ordinals of the form $\omega + n$ to label the odd numbers.

Definition A.3. A class X is *transitive* if for all $x \in X$ we have $x \subseteq X$, or equivalently, if for all x, y such that $y \in x \in X$ we have $y \in X$.

Theorem A.4. *Every natural number and the set of all natural numbers ω are transitive and well-ordered by the membership relation $\in$.*

From now on, we will denote the well-ordered set $(\omega, \in)$ as $(\omega, <)$ and write $n < m$ instead of $n \in m$ when referring to natural numbers. More generally, ordinal numbers are precisely the sets with the properties described by the previous theorem.

Definition A.5 (Ordinal number)**.** A set α is an *ordinal number* if it is transitive and well-ordered by the membership relation $\in$. We denote the *class of all ordinal numbers* by On.

Theorem A.6. *The class* On *is itself transitive and well-ordered by $\in$. This implies that it is not a set; otherwise,* $\mathrm{On} \in \mathrm{On}$*. Furthermore, if X is a transitive proper class well-ordered by $\in$, then $X = \mathrm{On}$.*

[3] The direction "$\Leftarrow$" actually cannot be proved in bare ZF, as we need some kind of choice principle to *choose* x_{i+1} from all the $x' \prec x_i$, and we need to do this infinitely many times.

As for notation, we will usually denote ordinals using letters from the beginning of the Greek alphabet: α, β, γ, $\delta, \ldots$ Furthermore, we compare ordinals using the symbol '$<$'. That is, we write $\beta < \alpha$ instead of $\beta \in \alpha$.

It can be shown that finite ordinals are exactly the natural numbers, and ω is the first infinite ordinal. Because ordinals are well-ordered, every set of ordinals $A \subseteq \mathrm{On}$ has a supremum $\sup A$, the least ordinal β such that $\beta \geq \alpha$ for all $\alpha \in A$.

Theorem A.7 (About types of well-orders)**.** *Every well-ordered set $(W, \preceq)$ is order-isomorphic to a unique ordinal number $(\alpha, \leq)$ known as the* order type *of W.*

Transfinite induction Ordinal numbers allow us to extend the concept of mathematical induction to *transfinite* induction on all ordinals.

If α is an ordinal, then we call all $\beta < \alpha$ the *predecessors* of α. The *successor* of α is the ordinal $\alpha + 1 \coloneqq \alpha \cup \{\alpha\}$. We say that α is the *direct predecessor* of $\alpha + 1$. It is easy to show that $\alpha + 1$ is the smallest ordinal larger than α.

Definition A.8. An ordinal number α is called

(a) an *isolated* ordinal if $\alpha = 0$ or if α has a direct predecessor,

(b) a *limit* ordinal otherwise.

Isolated ordinals $\alpha > 0$ are also sometimes called *successor* ordinals.

Example 2. Examples of isolated ordinals include all natural numbers or the ordinals $\omega + 1$, $\omega \cdot 2 + 7$, and $\omega^\omega + 2$, which we will define later. Examples of limit ordinals include ω, $\omega + \omega$, or $\omega \cdot \omega$.

Theorem A.9 (Transfinite Induction Principle)**.** *Let $\varphi(\alpha)$ be a property of ordinal numbers such that*

(i) *$\varphi(0)$ holds,* *... base case*

(ii) *whenever $\varphi(\alpha)$ holds, then $\varphi(\alpha + 1)$ holds as well,* *... successor stage*

(iii) *if α is a limit ordinal and $\varphi(\beta)$ holds for all $\beta < \alpha$, then $\varphi(\alpha)$ holds.*

Then $\varphi(\alpha)$ holds for every ordinal number α.

Proof. Suppose that φ fails for some ordinal γ, and let S be the set of all ordinals $\delta \leq \gamma$ for which φ fails. Since ordinals are well-ordered, the set S has a minimum element α: the first ordinal for which φ fails. From (i), $\alpha > 0$. If $\alpha = \beta + 1$ is isolated, then $\varphi(\alpha)$ holds by (ii) since $\varphi(\beta)$ holds. If α is limit, then $\varphi(\alpha)$ holds by (iii) since $\varphi(\beta)$ holds for all $\beta < \alpha$. Either way, we get a contradiction. □

One can similarly prove the following alternative formulation.

Theorem A.10 (Transfinite Induction Principle II)**.** *Let $\varphi(\alpha)$ be a property of ordinal numbers such that for every ordinal α, the following is true:*

if $\varphi(\beta)$ holds for all $\beta < \alpha$, then $\varphi(\alpha)$ holds as well.

Then $\varphi(\alpha)$ holds for every ordinal number α.

Ordinal arithmetic The definition above may not provide much intuition for working with ordinal numbers. We therefore briefly discuss basic ordinal arithmetic. Although this material will not be used later in the thesis, it may help readers unfamiliar with ordinals to develop some intuition.[4]

The *lexicographical ordering* of a Cartesian product $A \times B$ is defined by $(a_1, b_1) \leq_L (a_2, b_2)$ if $a_1 < a_2$, or $a_1 = a_2$ and $b_1 \leq b_2$. It is easy to see that if A and B are well-ordered by $\leq$, then $A \times B$ is well-ordered by $\leq_L$.

Definition A.11. Let α and β be ordinal numbers. We define ordinals

(a) $\alpha + \beta$ as the order type of the set $(\{0\} \times \alpha) \cup (\{1\} \times \beta)$ when ordered lexicographically,

(b) $\alpha \cdot \beta$ as the order type of the set $\beta \times \alpha$ when ordered lexicographically.

Notice that our previous notation of denoting $\alpha \cup \{\alpha\}$ by $\alpha + 1$ is consistent with the above definition. We can imagine $\alpha + \beta$ as a pile of decreasing matchsticks labeled by α, followed by another pile of matchsticks labeled by β. Notice that we are using $\beta \times \alpha$ in the definition of $\alpha \cdot \beta$. The ordinal $\alpha \cdot \beta$ can be imagined as taking a pile of matchsticks labeled by β, and then replacing each stick with a copy of α (a pile of matchsticks labeled by α).

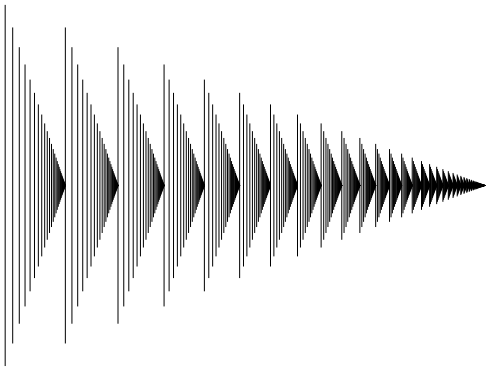


Figure 1: A representation of the ordinal $\omega \cdot \omega$. Each stick corresponds to an ordinal of the form $\omega \cdot m + n$ where m and n are natural numbers [GI15].

With this intuition, it should not be surprising that ordinal addition and multiplication are, in general, not commutative. It is easy to see that $1 + \omega = \omega$, but $\omega + 1 \neq \omega$. For multiplication, consider $2 \cdot \omega$, the order type of countably infinitely many copies of $\{0, 1\}$ stacked behind each other. This can be clearly labeled by ω, so $2 \cdot \omega = \omega$. But $\omega \cdot 2$ is the order type of two consecutive copies of ω. When we try to label them using ω, we use all $n \in \omega$ to label the first copy and need more ordinals for the second copy. Therefore $\omega \cdot 2 > \omega$.

Observation A.12. *For any ordinals α, β, γ and natural $n \in \omega$ it holds that*

(a) $\alpha + 0 = \alpha = 0 + \alpha, \qquad \alpha \cdot 0 = 0 = 0 \cdot \alpha, \qquad \alpha \cdot 1 = \alpha = 1 \cdot \alpha,$

(b) $\alpha + (\beta + \gamma) = (\alpha + \beta) + \gamma, \qquad \alpha \cdot (\beta \cdot \gamma) = (\alpha \cdot \beta) \cdot \gamma,$

(c) $\alpha \cdot 2 = \alpha + \alpha, \qquad \alpha \cdot 3 = \alpha + \alpha + \alpha, \qquad \alpha \cdot (n + 1) = \alpha \cdot n + \alpha.$

Definition A.13. For ordinal numbers α and β, we define α^β recursively as

(i) $\alpha^0 := 1$,

[4] I also recommend the following video by Vsauce, which illustrates how to construct larger ordinals from already existing ones: `https://www.youtube.com/watch?v=SrU9YDoXE88`.

(ii) $\alpha^{\beta+1} \coloneqq \alpha^\beta \cdot \alpha$,

(iii) if β is a limit ordinal, then $\alpha^\beta \coloneqq \sup\{\alpha^\gamma \mid 0 < \gamma < \beta\}$.

To get an intuition for ordinal powers, consider the ordinal $\omega^2 = \omega \cdot \omega$. It represents multiple copies of ω arranged in the same manner as ω. To construct $\omega^3 = (\omega \cdot \omega) \cdot \omega$, we take multiple copies of ω^2 and arrange them according to ω. If we repeat this process countably infinitely many times, we arrive at ω^ω. We can continue and arrive at larger and larger ordinals, such as $\omega^{(\omega^\omega)}$ or $\omega^{(\omega^{(\omega^\omega)})}$. Eventually, we construct the ordinal

$$\varepsilon_0 \coloneqq \sup\left\{\omega,\, \omega^\omega,\, \omega^{\omega^\omega},\, \omega^{\omega^{\omega^\omega}}, \dots\right\},$$

which is closely related to Peano Arithmetic. It might be surprising that this ordinal is still countable and, in a certain sense, quite small, even for countable ordinals. For more details, see the survey in Section 3 of [Smo26].

A.2 Cardinal Numbers

Similar to how ordinal numbers represent the order types of well-ordered sets, *cardinal numbers* represent the *sizes* of well-ordered sets.

Definition A.14. If x and y are sets, we define the following relations:

(a) $x \approx y$ if there exists a bijection $f\colon x \to y$, ... x and y are *equinumerous*

(b) $x \preceq y$ if there exists an injection $f\colon x \to y$, ... x is *subvalent* to y

(c) $x \prec y$ if $x \preceq y$ and $x \not\approx y$.

Theorem A.15 (Cantor's Theorem)**.** *For every set x, we have $x \prec \mathcal{P}(x)$.*

Theorem A.16 (Cantor–Bernstein Theorem)**.** $x \approx y \iff (x \preceq y \wedge y \preceq x)$.

Definition A.17 (Cardinal number)**.** An ordinal κ is a *cardinal number* if

$$(\forall \alpha \in \mathrm{On})(\alpha < \kappa \implies \alpha \prec \kappa).$$

We denote the *class of all cardinal numbers* by Cn.

Every $n \in \omega$ and ω are cardinal numbers. If $\alpha \geq \omega$, then $\alpha + 1$ is not a cardinal number; thus, every infinite cardinal number is a limit ordinal number. But not every limit ordinal number is a cardinal number: for instance $\omega + \omega > \omega$, but $\omega + \omega \approx \omega$, so $\omega + \omega$ is not a cardinal number. By the same logic, if $\alpha > \omega$ is countable, then it is not a cardinal number. On the other hand, the first uncountable ordinal ω_1 is a cardinal number. As for notation, we will denote cardinal numbers using letters from the middle of the Greek alphabet: κ, λ, μ, ν, ...

If a set x is equinumerous with a cardinal number κ, then we define $|x| \coloneqq \kappa$, and we say that κ is the *cardinality* of x. Observe that if α is an ordinal, then $|\alpha| \leq \alpha$, and that $|\alpha| = \alpha$ holds $\iff$ α is a cardinal number $\iff$ α is the first ordinal with cardinality $|\alpha|$.

Observation A.18. *If x and y have their cardinalities defined, then*

(a) $x \approx y \iff |x| = |y|$,

(b) $x \approx |x|$.

Observation A.19. *The cardinality $|x|$ is defined $\iff$ x can be well-ordered.*

Proof. If x can be well-ordered, then $|x|$ is the least order type of the well-orderings of x, as each of those orderings induces a bijection between x and the ordinal order type of the ordering. On the other hand, if $|x| = \kappa$ is defined, then we can well-order x by inheriting the order of κ. □

In this thesis, we always assume the axiom of choice, which is equivalent to the statement that all sets can be well-ordered. Hence, we can safely assume that every set has its cardinality defined and can thus be replaced by an equinumerous cardinal number.

Alephs Cardinal numbers are closed under taking suprema: If $\lambda = \sup \kappa_i$ were not a cardinal, then $|\lambda| < \lambda$. Since λ is the supremum, there is some κ_j such that $|\lambda| < \kappa_j \leq \lambda$. But this contradicts κ_j being a cardinal. One can also show that for every cardinal, there exists a larger cardinal. Therefore, the class of all cardinals Cn is a proper class; otherwise its supremum would be the largest cardinal.

From the fact that Cn is a closed proper class of ordinals, one can prove that there is a uniquely determined bijective increasing mapping $\aleph\colon \mathrm{On} \to \mathrm{Cn} \setminus \omega$ enumerating infinite cardinals that satisfies $\aleph(\alpha) = \sup\{\aleph(\beta) \mid \beta < \alpha\}$ for every limit ordinal α. Cantor introduced the symbol $\aleph$ ("aleph"), the first letter of the Hebrew alphabet, to denote this function; its values $\aleph(\alpha)$ are denoted by $\aleph_\alpha$.

Historically, ordinals and cardinals were not concrete sets but abstract concepts: ordinals described well-ordering types, while cardinals measured size. This distinction led to the development of two parallel notation systems, ω_α and $\aleph_\alpha$. Von Neumann's 1923 set-theoretic definition of ordinals unified these ideas by providing canonical representatives for ordinal types. Today, we often write ω_α when considering the cardinal $\aleph_\alpha$ viewed as an ordinal with its well-order. In particular, the first uncountable ordinal is denoted by ω_1.

The *successor* of a cardinal κ is the smallest cardinal larger than κ, and we denote it by κ^+. Furthermore, we say that κ is the *predecessor* of κ^+. Finally, $\lambda > 0$ is a *limit* cardinal if it has no predecessor.

Example 3. Clearly $\aleph_{\alpha+1} = \aleph_\alpha^+$ are successor cardinals, and $\aleph_0$ is a limit cardinal. It is easy to show that $\aleph_\alpha$ for $\alpha > 0$ is a limit cardinal $\iff$ α is a limit ordinal.

Cardinal arithmetic If κ and λ are cardinal numbers, we define cardinals

(a) $\kappa + \lambda \coloneqq |(\{0\} \times \kappa) \cup (\{1\} \times \lambda)|$,

(b) $\kappa \cdot \lambda \coloneqq |\lambda \times \kappa|$,

(c) $\kappa^\lambda \coloneqq |\{f \mid f\colon \lambda \to \kappa\}|$.

In other words, $\kappa + \lambda$ and $\kappa \cdot \lambda$ are cardinal numbers that represent the size of the set on the right side of the equation, while ordinal addition and multiplication express the order type of the same set when ordered lexicographically. Observe that cardinal addition and multiplication are associative, commutative, and distributive. Ordinal addition and multiplication are associative, but in general, they are not commutative or right-distributive. Intuitively, this is because the ordinal operations "keep track" of the underlying orderings. Also note that $2^\lambda = |\mathcal{P}(\lambda)|$ as there is a clear 1-to-1 correspondence between functions $f\colon \lambda \to \{0, 1\}$ and subsets $A \subseteq \lambda$.

Theorem A.20. *For every ordinal α, we have $|\aleph_\alpha \times \aleph_\alpha| = \aleph_\alpha$.*

From this, it is not difficult to show that if κ and λ are cardinals, and at least one of them is infinite, then $\kappa + \lambda = \max\{\kappa, \lambda\}$. If, in addition, they are nonzero, then $\kappa \cdot \lambda = \max\{\kappa, \lambda\}$. When applied to infinite sets, we get:

Corollary A.21. *If A and B are infinite sets, then*

$$|A \cup B| = |A \times B| = \max\big\{|A|, |B|\big\}.$$

Furthermore, if $|A| > |B|$, then $|A \setminus B| = |A|$.

Lemma A.22. *For any set S it holds that $|\bigcup S| \leq |S| \cdot \sup\{|A| \mid A \in S\}$.*

Corollary A.23. *The union of an arbitrary collection of $\aleph_\alpha$ sets, each of cardinality at most $\aleph_\alpha$, has cardinality at most $\aleph_\alpha$.*

Regular cardinals The pigeonhole principle states that ω cannot be partitioned into a finite number of finite sets, or equivalently, that if $A \subseteq \omega$ is finite, then $\sup(A) < \omega$. Hence, we cannot "reach" $\aleph_0 = \omega$ from below via any subset of smaller cardinality. Infinite cardinals with this property are called *regular*.

Definition A.24 (Cofinality)**.** A subset $A \subseteq \alpha$ of a limit ordinal α is *cofinal* in α if $\sup(A) = \alpha$ (the limit of the increasing sequence of elements of A is α). The *cofinality* of α is the "length" of the shortest increasing sequence with limit α:

$$\operatorname{cf}(\alpha) \coloneqq \min\{\text{order type of } (A, \leq) \mid A \subseteq \alpha \wedge \sup(A) = \alpha\}.$$

It can be shown that $\operatorname{cf}(\alpha)$ is equal to the minimal size $|A|$ of a cofinal subset A of α, so cofinality is always an infinite cardinal. Therefore $\omega \leq \operatorname{cf}(\alpha) \leq |\alpha|$.

Definition A.25 (Regular cardinal)**.** An infinite cardinal number κ is a *regular cardinal* if $\operatorname{cf}(\kappa) = \kappa$, and a *singular cardinal* otherwise.

Regular cardinals behave similarly to ω in that they are *almost* closed under taking suprema: as long as the length of the sequence is less than κ, the limit will never reach κ. We can equivalently express this via unions of smaller sets.

Theorem A.26. *An infinite cardinal number κ is singular $\iff$ there exists a set X such that $\kappa = \bigcup X$, where $|X| < \kappa$ and $|x| < \kappa$ for all $x \in X$.*

It is not difficult to show that $\operatorname{cf}(\operatorname{cf}(\alpha)) = \operatorname{cf}(\alpha)$, so cofinality is always a regular cardinal. We know that ω is regular, and it can be shown in ZFC that every infinite successor cardinal $\aleph_{\alpha+1}$ is also regular. Perhaps somewhat surprisingly, the existence of a regular limit cardinal other than ω cannot be proved from the axioms of ZFC because it would allow one to construct a model of ZFC inside ZFC, contradicting Gödel's second incompleteness theorem. These hypothetical cardinals were first suggested by Felix Hausdorff in 1908, and today we call them *weakly inaccessible*. If a weakly inaccessible cardinal κ further satisfies that for all cardinals $\lambda < \kappa$ we have $2^\lambda = |\mathcal{P}(\lambda)| < \kappa$, then κ is said to be *strongly inaccessible*, or often simply just *inaccessible*.

Large cardinals Many problems in set theory and infinite combinatorics lead to the question of whether cardinals with certain properties exist. When the existence of such cardinals cannot be proved in ZFC, they are called *large cardinals*. Statements asserting the existence of such cardinals are known as *large cardinal axioms*, and they form a well-studied hierarchy [Kan94]. The existence of weakly and strongly inaccessible cardinals are the weakest of these axioms.

Some theorems can be proved only under assumptions asserting the existence of appropriate large cardinals. One has to take a "leap of faith" when making such assumptions since ZFC cannot prove their consistency.

A.3 Filters and Ultrafilters

Throughout this section, X is always a nonempty set, and given a set $A \subseteq X$, we denote by $\overline{A}$ the set $X \setminus A$. For a nonempty system of subsets $\mathcal{A} \subseteq \mathcal{P}(X)$, we denote by $\mathcal{A}^*$ the set $\{\overline{A} \mid A \in \mathcal{A}\}$.

Definition A.27. Let $X \neq \varnothing$. A nonempty system of subsets $\mathcal{F} \subseteq \mathcal{P}(X)$ is a *filter* on X if

(i) $\varnothing \notin \mathcal{F}$,

(ii) if $A \in \mathcal{F}$ and $A \subseteq B$, then $B \in \mathcal{F}$, ... $\mathcal{F}$ is *upward closed*

(iii) if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

A filter $\mathcal{F}$ that also satisfies

(iv) $\forall A \subseteq X$ either $A \in \mathcal{F}$ or $\overline{A} \in \mathcal{F}$

is called an *ultrafilter* on X. Given an ultrafilter $\mathcal{U}$ on X, we say that a subset $A \subseteq X$ is *$\mathcal{U}$-large* if $A \in \mathcal{U}$.

Intuition. A filter on X allows us to express which subsets of X we consider to be "large." (i) The empty set is not large. (ii) Adding something to a large set will not make it small. (iii) An intersection of two large sets is large. (iv) Whenever we split X into two parts, one of them has to be large.

Observation A.28. *For every filter $\mathcal{F}$ on X it holds that*

(a) *condition* (i) *is equivalent to* $\mathcal{F} \neq \mathcal{P}(X)$, *... from* (ii)

(b) $X \in \mathcal{F}$, *... from* $\mathcal{F} \neq \varnothing$ *and* (ii)

(c) $\forall A, B \in \mathcal{F} : A \cap B \neq \varnothing$, *... from* (iii) *and* (i)

(d) $\mathcal{F}$ *is closed under finite intersections, ... from* (iii)

(e) $\nexists A \subseteq X$ *such that* $A, \overline{A} \in \mathcal{F}$, *... from* (iii) *and* (i)

The last observation implies that the "or" in (iv) *is exclusive.*

Lemma A.29. *If $\mathcal{U}$ is an ultrafilter on X and $A = A_1 \cup \cdots \cup A_k \in \mathcal{U}$, then $A_j \in \mathcal{U}$ for at least one index j. Furthermore, if the sets A_i are disjoint, then the index j is unique.*

Proof. Suppose that $A_j \notin \mathcal{U}$ for all j. Then from (iv) we have $\overline{A}_j \in \mathcal{U}$ for all j and (iii) implies that $\bigcap \overline{A}_j = \overline{A} \in \mathcal{U}$. Hence, both A and $\overline{A}$ are in $\mathcal{U}$, a contradiction. Notice that if the sets A_i are disjoint, then (iii) and (i) further imply that the index j must be unique. □

Definition A.30 (Maximal filter)**.** A filter $\mathcal{F}$ on X is *maximal* if there is no filter $\mathcal{F}'$ on X such that $\mathcal{F} \subset \mathcal{F}'$.

Theorem A.31. *For a filter $\mathcal{F}$ on X, the following conditions are equivalent:*

(i) $\mathcal{F}$ *is an ultrafilter,*

(ii) $\mathcal{F}$ *is a maximal filter,*

(iii) *whenever $A \cup B \in \mathcal{F}$, then $A \in \mathcal{F}$ or $B \in \mathcal{F}$.*

Proof. We already know (i)⇒(iii) from Lemma A.29.

(iii)⇒(ii) Suppose $\mathcal{F} \subset \mathcal{F}'$ for some filter $\mathcal{F}'$. Then there is some $A \in \mathcal{F}'$ such that $A \notin \mathcal{F}$, hence $\overline{A} \in \mathcal{F}$ since $A \cup \overline{A} = X \in \mathcal{F}$. Because $\mathcal{F} \subseteq \mathcal{F}'$, also $\overline{A} \in \mathcal{F}'$, so both A and $\overline{A}$ are in $\mathcal{F}'$, a contradiction.

(ii)⇒(i) Suppose that $\mathcal{F}$ is not an ultrafilter. Then there exists a set $A \subseteq X$ such that $A, \overline{A} \notin \mathcal{F}$. Note that for all $B \in \mathcal{F}$ we have $A \cap B \neq \varnothing$; otherwise $B \subseteq \overline{A}$, so we would have $\overline{A} \in \mathcal{F}$. Define a filter

$$\mathcal{F}' \coloneqq \{B' \subseteq X \mid (\exists B \in \mathcal{F}) : (A \cap B) \subseteq B'\}.$$

It is indeed a filter: $\varnothing \notin \mathcal{F}'$ since $A \cap B \neq \varnothing$ for all B; it is upward closed from definition; and it is intersection closed because if $B'_1, B'_2 \in \mathcal{F}'$, then $A \cap B_1 \subseteq B'_1$ and $A \cap B_2 \subseteq B'_2$ for some $B_1, B_2 \in \mathcal{F}$. Since $B_1 \cap B_2 \in \mathcal{F}$, we have $A \cap B_1 \cap B_2 \subseteq B'_1 \cap B'_2 \in \mathcal{F}'$.

Notice that $\mathcal{F}'$ is a filter such that $\mathcal{F} \subseteq \mathcal{F}'$ and $A \in \mathcal{F}'$ but $A \notin \mathcal{F}$, so $\mathcal{F}$ is not maximal. □

Two important examples

- For every set X and every nonempty $Y \subseteq X$, the set

$$\mathcal{F}_Y \coloneqq \{A \subseteq X \mid Y \subseteq A\}$$

  is a filter on X. Filters of this form are called *principal filters*.

- The filter

$$\mathcal{F}_{co} \coloneqq \{A \subseteq X \mid \overline{A} \text{ is finite}\}$$

  is the *cofinite filter* on X.

Lemma A.32. *A principle filter $\mathcal{F}_Y$ on X is an ultrafilter $\iff$ $|Y| = 1$.*

Proof. "⇒" Suppose that $x, y \in Y$ and $x \neq y$. Then $Y = (Y \setminus \{x\}) \cup \{x\} \in \mathcal{F}_Y$, but neither $Y \setminus \{x\}$ nor $\{x\}$ are in $\mathcal{F}_Y$, contradicting the previous lemma.

"⇐" We know that $\mathcal{F}_Y$ is a filter. To show that it is an ultrafilter, let $Y = \{x\}$ be a singleton and suppose that $A \subseteq X$. Either $x \in A$, in which case $A \in \mathcal{F}$, or $x \notin A$, in which case $x \in \overline{A}$, so $\overline{A} \in \mathcal{F}$. □

Definition A.33 (Nonprincipal ultrafilter)**.** Ultrafilters of the form

$$\mathcal{U}_x = \{A \subseteq X \mid x \in A\}$$

for some $x \in X$ are called *principal*. Ultrafilters that are not principal are called *nonprincipal* or *free*, and they are the interesting ones.

Lemma A.34. *For an ultrafilter $\mathcal{U}$ on X, the following conditions are equivalent:*

(i) *$\mathcal{U}$ is a principal ultrafilter,*

(ii) *$\mathcal{U}$ contains a finite set,*

(iii) $\bigcap \mathcal{U} = \bigcap_{A \in \mathcal{U}} A \neq \varnothing$.

In particular, it follows from (ii) *that all ultrafilters on a finite set X are principal.*

Proof. (i)⇒(iii) If $\mathcal{U} = \{A \subseteq X \mid x \in A\}$ for some $x \in X$, then $x \in \bigcap \mathcal{U}$.

(iii)⇒(ii) Let $A_0 = \bigcap \mathcal{U}$ and note that $\overline{A_0} \notin \mathcal{U}$ (since $A_0 \subseteq A$ for all $A \in \mathcal{U}$), so $A_0 \in \mathcal{U}$ because $\mathcal{U}$ is an ultrafilter. We claim that $|A_0| = 1$. Suppose $|A_0| > 1$, so A_0 has a proper nonempty subset $B \subseteq A_0$. But then neither B nor $\overline{B}$ contains A_0 as a subset, so $B, \overline{B} \notin \mathcal{U}$, a contradiction with $\mathcal{U}$ being an ultrafilter.

(ii)⇒(i) Let $A_0 \in \mathcal{U}$ be a set of the smallest (finite) size.

If $A_0 = \{x\}$ is a singleton, then for all $B \in \mathcal{U}$ we have $x \in B$, since otherwise $B \cap A_0 = \varnothing$. Hence $\mathcal{U} \subseteq \mathcal{U}_x = \{A \subseteq X \mid x \in A\}$. To show equality, suppose that $x \in A$ but $A \notin \mathcal{U}$. Then $\overline{A} \in \mathcal{U}$ since $\mathcal{U}$ is an ultrafilter, but $x \notin \overline{A}$.

If $|A_0| > 1$, then A_0 contains a proper nonempty subset $B \subset A_0$. Because $\mathcal{U}$ is an ultrafilter and $B \notin \mathcal{U}$ (since A_0 has the smallest size in $\mathcal{U}$), we have $\overline{B} \in \mathcal{U}$. Hence $\overline{B} \cap A_0 \in \mathcal{U}$, but $|\overline{B} \cap A_0| < |A_0|$, which contradicts the minimality of A_0 as well. □

Corollary A.35. *An ultrafilter $\mathcal{U}$ on an infinite set X is nonprincipal $\iff$ it contains the cofinite filter $\mathcal{F}_{co}$.*

Proof. If an ultrafilter is nonprincipal, then it contains no finite sets, so it must contain all their complements. On the other hand, if it $\overline{A} \in \mathcal{U}$ for every finite $A \subseteq X$, then also $A \notin \mathcal{U}$ for every finite set $A \subseteq X$, so $\mathcal{U}$ contains no finite sets, and thus is a principal ultrafilter. □

Theorem A.36 (AC, The Ultrafilter Lemma, Tarski, 1930)**.** *Every filter $\mathcal{F}$ on a set X can be extended to an ultrafilter $\mathcal{U} \supseteq \mathcal{F}$ on X.*

Proof. We will use Zorn's lemma. Let $\mathbf{F} \subseteq \mathcal{P}(\mathcal{P}(X))$ be the set of all filters on X. Notice that if $\mathcal{C}$ is a chain of filters in the partially ordered set $(\mathbf{F}, \subseteq)$, then $\bigcup \mathcal{C} = \bigcup_{\mathcal{F} \in \mathcal{C}} \mathcal{F}$ is also a filter and an upper bound of $\mathcal{C}$. Hence, we can apply Zorn's lemma to obtain a $\subseteq$-maximal filter $\mathcal{F}'$ above $\mathcal{F}$. Theorem A.31 implies that $\mathcal{F}'$ is an ultrafilter. □

Corollary A.37 (AC)**.** *Every infinite set has a nonprincipal ultrafilter.*

Proof. Use the previous theorem to extend the cofinite filter. □

This result can be further strengthened.

Definition A.38 (Uniform filter)**.** A filter $\mathcal{F}$ on an infinite set X is *uniform* if all $A \in \mathcal{F}$ have the same cardinality as X.

Observation A.39. *Every uniform ultrafilter is nonprincipal, as it contains no finite sets.*

Proposition A.40 (AC)**.** *Every infinite set has a uniform ultrafilter.*

We do this by extending a more powerful filter than the cofinite filter. For an infinite set X with $|X| = \kappa$, we call the filter

$$\mathcal{F} \coloneqq [X]^{<\kappa} = \{A \subseteq X \mid |X \setminus A| < \kappa\}$$

the *Fréchet filter* on X. The Fréchet filter on a countably infinite set X is just the cofinite filter.

Proof of Proposition A.40. Let X be an infinite set and let $\mathcal{U}$ be an ultrafilter that extends the Fréchet filter $\mathcal{F}$ on X. Suppose that $A \in \mathcal{U}$ has size $|A| < \kappa$. Then $\overline{A} \in \mathcal{F} \subseteq \mathcal{U}$, a contradiction with $\mathcal{U}$ being an ultrafilter. □

Definition A.41 (κ-complete filter)**.** A filter $\mathcal{F}$ on a set X is *κ-complete* if any intersection of less than κ sets from $\mathcal{F}$ is still in $\mathcal{F}$.

Observation A.42. *If $\mathcal{U}$ is a κ-complete ultrafilter and $A = \bigcup \langle A_\alpha \mid \alpha < \lambda \rangle \in \mathcal{U}$ where $\lambda < \kappa$, then $A_\gamma \in \mathcal{U}$ for some index γ. Furthermore, if the sets A_α are disjoint, then this γ is unique.*

Proof. The same as Lemma A.29. □

Every filter is ω-complete. If a filter is ω_1-complete, then it is closed under countable intersections. These filters have been extensively studied and are often referred to as σ-complete instead. For example, the filter on $\mathbb{R}$ consisting of sets $A \subseteq \mathbb{R}$ such that $\mathbb{R} \setminus A$ has Lebesgue measure zero is σ-complete (since the Lebesgue measure is countably additive). Notice that every principal filter $\mathcal{F}_Y$ is κ-complete for all cardinals κ because its intersection $\bigcap \mathcal{F}_Y = Y$ is nonempty. The Fréchet filter on a cardinal λ is $\mathrm{cf}(\lambda)$-complete, but not $\mathrm{cf}(\lambda)^+$-complete.

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